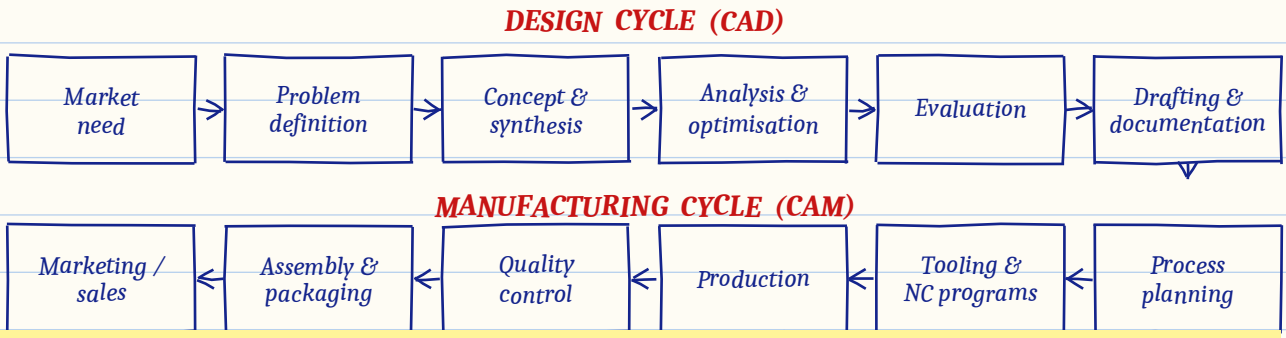
UNIT – I : Overview of CAD/CAM & Computer Graphics

1. Product Cycle

- **Product cycle** = all the activities from the customer's need / idea of a product till it reaches the customer (and its disposal).
- It is divided into two halves:
  - **Design cycle** (office / CAD side): market need → problem definition → conceptual design → analysis & optimisation → evaluation → drawings & documentation.
  - **Manufacturing cycle** (shop-floor / CAM side): process planning → tooling & NC programs → production → quality control → assembly → packaging → marketing & sales.
- In the conventional (manual) cycle the two halves are separate, so information is re-typed, redrawn and misunderstood → long lead time and errors.
- With computers a **common database** is shared by both halves, so the cycle is shorter and cheaper.



Concurrent / simultaneous engineering: design and manufacturing teams work in parallel on the same data instead of one after the other.

2. CAD – Computer Aided Design

- **CAD** is the use of computer systems to assist in the creation, modification, analysis or optimisation of a design. (Groover)
- Hardware: computer (CPU), graphics display (CRT / LCD), input devices (keyboard, mouse, light pen, digitiser, scanner), output devices (plotter, printer).

- Software: modelling package + analysis package + database.

Design process and where CAD helps (Shigley's 4 phases)

- **Geometric modelling** – describe the shape mathematically (wire-frame, surface, solid).
- **Engineering analysis** – FEA, stress, heat transfer, mass properties, kinematics, vibration.
- **Design review & evaluation** – interference checking, animation, simulation, tolerance check.
- **Automated drafting** – 2D views, sections, dimensioning, bill of materials, drawing standards.

Benefits of CAD

- Higher designer productivity (3 to 10 times) and shorter design time.
- Better design quality – more alternatives can be analysed.
- Easy modification, standardised drawings, better communication.
- Creates a manufacturing database (geometry, dimensions, BOM, tolerances) that CAM can reuse directly.

3. CAM – Computer Aided Manufacturing

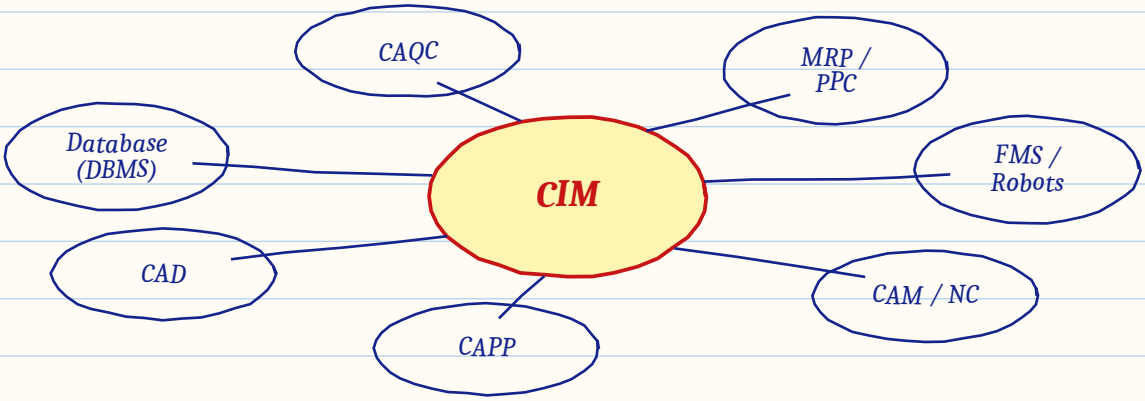
- **CAM** = use of computer systems to plan, manage and control the operations of a manufacturing plant through direct or indirect computer interface with the plant's production resources.
- Two broad areas:
  - **Computer monitoring and control** – the computer is connected directly to the process (NC / CNC machines, robots, process control).
  - **Manufacturing support applications** – the computer is used indirectly: process planning, part programming, production scheduling, inventory, MRP, shop-floor control.
- Examples: NC part programming, CNC & DNC, robot programming, automated inspection (CMM), AGV and AS/RS, FMS.

Benefits of CAM

- Reduced lead time, scrap and rework; consistent quality.
- Better machine utilisation, flexible production, lower inventory.
- Less skilled-labour dependence, better safety.

4. CIM – Computer Integrated Manufacturing

- **CIM** = integration of all business and manufacturing functions (design, planning, production, quality, marketing, accounts) through a **common computer database** and network. (CASA/SME wheel)
- CIM = CAD + CAM + business functions (production planning & control, MRP, costing, sales).
- Goal: "right information at the right place at the right time" → agile, low-cost, high-quality production.



CAD/CAM vs CIM

Feature	CAD / CAM	CIM
Scope	design + manufacturing only	whole enterprise incl. business & marketing
Data	may be separate islands	one shared database
Aim	automate individual tasks	integrate all tasks
Benefit	faster design & production	shortest lead time, max flexibility

5. CAD Tools and CAM Tools

CAD tools (software / functions)

- **Drafting tools**: AutoCAD, DraftSight – 2D drawing, layers, dimensioning, hatching.
- **3D modelling tools**: CATIA, Creo (Pro/E), SolidWorks, NX (Unigraphics), Autodesk Inventor.
- **Analysis tools**: ANSYS, Abaqus, NASTRAN – FEA; Adams – mechanisms.
- **Utilities**: data exchange (IGES, STEP, DXF), rendering, animation, tolerance analysis, PDM.

CAM tools

- **NC programming systems**: Mastercam, NX CAM, PowerMill, Fusion 360 CAM, CATIA machining, APT.
- **Process planning (CAPP) software**, tool-path simulation and verification, post-processors.
- **Shop-floor tools**: DNC software, MES, MRP/ERP packages, CMM and inspection software, robot simulators.

6. Utilisation of CAD/CAM in an Industrial Environment

- Aerospace and automobile – body surface design, crash analysis, die & mould making.
- Machine-tool and capital goods – design of parts, assemblies and NC machining of complex parts.
- Electronics – PCB layout, chip design (VLSI), automated assembly.
- Architecture & construction (CAAD), shipbuilding, plant layout, piping design.
- Tool, die & mould shops – 3 to 5 axis machining from CAD solid models.
- Typical implementation steps in a company:
  - study present system and set objectives → select hardware / software (use evaluation criteria)
  - plan installation & training → pilot project → full implementation → review benefits.

7. Evaluation Criteria for a CAD/CAM System

- **Functional**: modelling capability (2D / 3D / solid / surface), drafting, analysis, NC module, data exchange (IGES / STEP).
- **Hardware**: CPU speed, RAM, graphics card, display resolution, storage, plotter.
- **Software**: user-friendliness (GUI), customisation (macros, API, programming language), database, vendor support.
- **Cost**: initial cost + training + maintenance + upgrades (total life-cycle cost).
- **Compatibility** with existing systems, networking ability, expansion / upgrade path.
- **Vendor**: reputation, service, documentation, user base and training.
- **Performance**: response time, regeneration speed, accuracy, reliability.
- Final decision usually by a **weighted-score (rating) method** – each criterion given a

weight, each system scored, total compared.

8. CAD Data Structure

- **Data structure** = the way geometric and non-geometric data of a model are organised in computer memory so they can be stored, searched and modified efficiently.
- Information stored for a model:
  - **Geometric data** – coordinates of points, equations of lines / curves / surfaces.
  - **Topological data** – how entities connect (vertex – edge – face relations).
  - **Non-graphic data** – material, colour, layer, dimensions, tolerances, weight, cost.

Common structures

- **Array / table** – fixed list of coordinates; simple but not flexible.
- **Linked list** – each record holds data + a pointer to the next; easy insertion / deletion.
- **Tree (hierarchical)** – parent–child structure, e.g. assembly → sub-assembly → part → feature (CSG tree).
- **Network / graph** – nodes joined in many-to-many manner (B-rep: V, E, F).
- **Relational tables** – data in rows and columns joined by keys.
- A good structure should give fast access, easy modification, small memory and data integrity.

9. Database Management System (DBMS)

- **Database** = organised collection of related data stored once and shared by many users / programs (CAD, CAM, CAPP, MRP). **DBMS** = software that creates, stores, retrieves, updates and protects it.
- Functions: data definition, storage, retrieval, security, integrity, backup & recovery, multi-user concurrency control.
- Advantages: reduced redundancy, data consistency, sharing, standards, security, data independence.

Database models

Model	Structure	Remark
Hierarchical	tree: one parent, many children	fast, rigid; assemblies / BOM
Network	many-to-many links (pointers)	complex but flexible
Relational	tables of rows & columns joined by keys	most popular; SQL queries
Object-oriented	data + methods in objects	suits CAD features, complex geometry

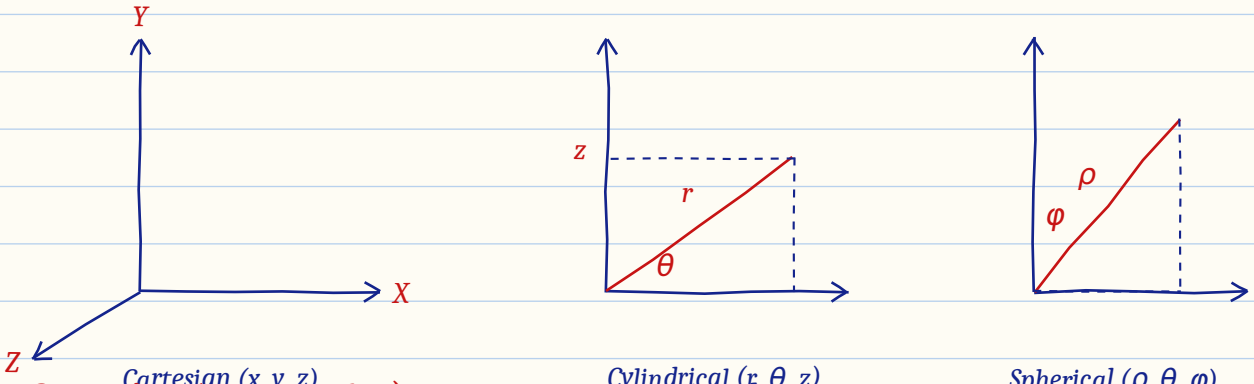
- **Schema** = overall design of the database; **sub-schema** = a user's view; **DDL** defines data, **DML** manipulates data.
- Four requirements for CAD databases: handle complex geometry, version control, long transactions, and links to graphic & non-graphic data.

Computer Graphics

10. Coordinate Systems

- A coordinate system gives numbers (coordinates) to the position of a point in space.
- Three systems are used in a graphics package:
  - **Model (world / master) coordinate system (MCS / WCS)** – the part is defined in real-world units (mm); origin chosen by designer.
  - **Working coordinate system (WCS)** – a temporary user-defined system, placed at convenient position / orientation (e.g. on a face) to draw 2D features on it.
  - **Screen (device) coordinate system (SCS)** – display pixels, origin at screen corner; units are pixels.
- Mapping: world → (viewing transformation) → normalised device coordinates (0 to 1) → (workstation transform) → screen pixels.

Types of coordinate systems in space



Cartesian (x, y, z) Cylindrical (r, θ, z) Spherical (ρ, θ, φ)

- **Cartesian (rectangular)**: x, y, z mutually perpendicular. Right-handed system is standard.
- **Cylindrical**: $x = r\cos\theta$, $y = r\sin\theta$, $z = z$.
- **Spherical**: $x = \rho\sin\varphi\cos\theta$, $y = \rho\sin\varphi\sin\theta$, $z = \rho\cos\varphi$.
- **Polar** (2D): (r, θ) – used in NC programming for circular patterns.

11. Graphics Package Functions

- Graphics package = software library to create, display and manipulate pictures (e.g. GKS, PHIGS, OpenGL, DirectX).
- Standard graphics (GKS – Graphical Kernel System; PHIGS) give device independence and portability.
- Main functions:
  - **Output primitives**: point, line, polyline, circle, arc, text, polygon fill – to generate the picture.
  - **Attributes**: colour, line type & width, text size & font, fill style.
  - **Geometric transformations**: translation, scaling, rotation, reflection.
  - **Viewing functions**: window, viewport, zoom, pan, projections.
  - **Segmenting / structuring**: grouping primitives into named pictures that can be edited.
  - **Input functions**: locator, pick, choice, valuator, string, stroke (interactive devices).
  - **Control & clipping**: initialise workstation, clear screen, clipping against window.
  - **Hidden line / surface removal, shading, colouring** for 3D realism.

Window and viewport

- **Window** = region of the world (model) space to be displayed. **Viewport** = region of the screen where it is shown.

$$xv = xvmin + (xw - xwmin)\cdot sx\,,\quad yv = yvmin + (yw - ywmin)\cdot sy$$

- Scale factors $sx = (xvmax - xvmin)/(xwmax - xwmin)$, $sy = (yvmax - yvmin)/(ywmax - ywmin)$.
- Zooming = shrinking the window; panning = moving the window.

12. Transformations – why needed

- Transformations change the position, size or orientation of an object **without redrawing** it.
- Basic: **translation, scaling, rotation** (+ reflection, shear). Used in modelling, assembly, animation and viewing.
- All are done by multiplying the point coordinates by a matrix. To treat translation also as multiplication we use **homogeneous coordinates**: a point (x, y) is written as $(x, y, 1)$ and a 3D point as $(x, y, z, 1)$.

13. 2D Transformations (homogeneous 3 × 3 matrices)

(a) Translation – move by (tx, ty)

$$x' = x + tx\,,\quad y' = y + ty$$

$$[x'\ y'\ 1] = [x\ y\ 1]\cdot \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ tx & ty & 1 \end{bmatrix}$$

(b) Scaling – by sx, sy about the origin

$$x' = sx\cdot x\,,\quad y' = sy\cdot y$$

$$[S] = \begin{bmatrix} sx & 0 & 0 \\ 0 & sy & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$sx = sy \rightarrow$ uniform scaling; $sx \neq sy \rightarrow$ differential scaling.

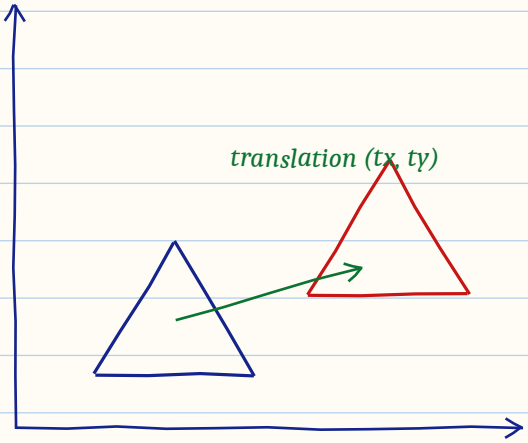
- Scaling about a fixed point (xf, yf): translate (–xf, –yf) → scale → translate (xf, yf).

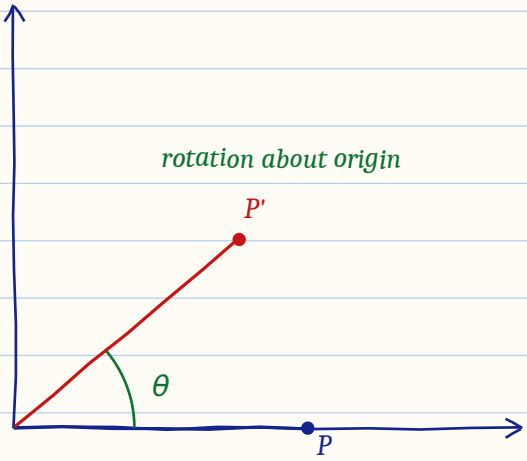
(c) Rotation – by angle θ (anticlockwise +ve) about the origin

$$x' = x\cos\theta - y\sin\theta \;,\; y' = x\sin\theta + y\cos\theta$$

$$[R] = \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

- Rotation about any point (h, k): $T(-h,-k)\cdot R(\theta)\cdot T(h,k)$ (three steps – concatenation).





(d) Reflection (mirror)

- About x-axis: x' = x, y' = –y. About y-axis: x' = –x, y' = y. About origin: both signs change.
- About line y = x: x' = y, y' = x. (the matrix has 1's on the anti-diagonal.)

(e) Shear

- x-shear: x' = x + shxy, y' = y. y-shear: y' = y + shyx, x' = x.

Concatenation of transformations

- Several transformations can be combined into one matrix: M = M1·M2·M3 … (**order matters**, matrix multiplication is not commutative: TR ≠ RT).

Solved example idea: rotate a triangle A(1,1) B(4,1) C(2,3) by 90° about origin → A'(–1,1) B'(–1,4) C'(–3,2).

14. 3D Transformations (4 × 4 homogeneous matrices)

- Point (x, y, z) → (x, y, z, 1); transformation: [x' y' z' 1] = [x y z 1] ·[T].

Translation

$$[T] = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ tx & ty & tz & 1 \end{bmatrix}$$

Scaling

$$[S] = \begin{bmatrix} sx & 0 & 0 & 0 \\ 0 & sy & 0 & 0 \\ 0 & 0 & sz & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Rotation about the axes (right-hand rule)

- **About z-axis** (θ): $x' = x\cos\theta - y\sin\theta$, $y' = x\sin\theta + y\cos\theta$, $z' = z$.
- **About x-axis** (θ): $y' = y\cos\theta - z\sin\theta$, $z' = y\sin\theta + z\cos\theta$, $x' = x$.
- **About y-axis** (θ): $z' = z\cos\theta - x\sin\theta$, $x' = z\sin\theta + x\cos\theta$, $y' = y$.
- Rotation about an arbitrary axis: translate the axis to the origin → rotate it to coincide with a coordinate axis (two rotations) → rotate by θ → reverse the two rotations → translate back.

Reflection in 3D

- About xy-plane: z → –z; about yz-plane: x → –x; about zx-plane: y → –y.

Projections (3D → 2D screen)

- **Parallel**: projectors parallel – orthographic (front, top, side views) and axonometric (isometric etc.). Keeps true size & parallelism.
- **Perspective**: projectors meet at the centre of projection (eye) – gives realistic view; far objects look smaller; one-, two-, three-point perspective.

15. Clipping

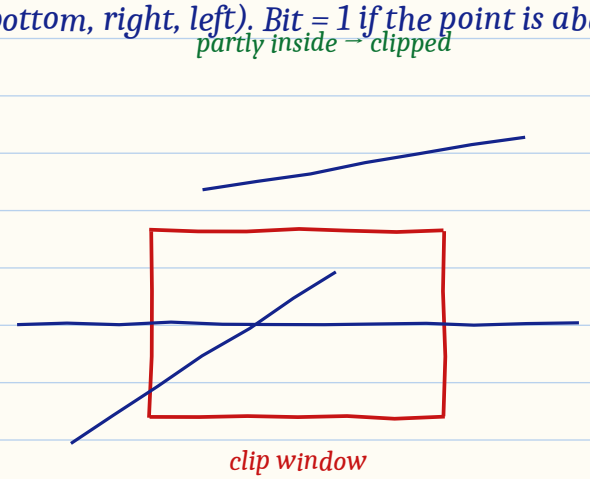
- **Clipping** = removing the parts of the picture lying outside the window (or view volume) so only the visible part is displayed / processed.
- Needed to save computing time and avoid drawing outside the screen area.
- Types: point clipping, line clipping, polygon clipping, text clipping, 3D clipping.

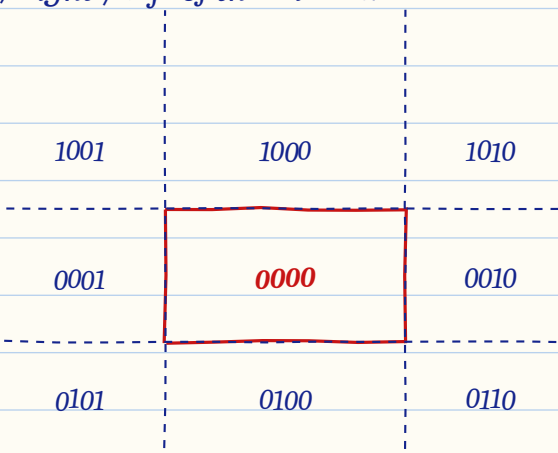
Point clipping

- A point (x, y) is visible if xwmin ≤ x ≤ xwmax and ywmin ≤ y ≤ ywmax.

Cohen–Sutherland line clipping algorithm

- Divide the plane into 9 regions; give each end-point a **4-bit outcode** (bit order: top, bottom, right, left). Bit = 1 if the point is above / below / right / left of the window.





- Steps: (1) find outcodes of both ends P1, P2.
  - If both codes are 0000 → line fully **inside** → **accept**.
  - If (code1 AND code2) ≠ 0000 → fully **outside** on same side → **reject**.
  - Otherwise line is partly inside: pick an outside end, find its intersection with a window edge, replace the end with it, recompute outcode and repeat.
- Intersection formulas (slope m = (y2 − y1)/(x2 − x1)): with left/right edge x = xw → y = y1 + m(xw − x1); with top/bottom edge y = yw → x = x1 + (yw − y1)/m.

Other methods

- **Midpoint subdivision** – repeatedly divide the line at its midpoint and test each half (binary search for intersection).
- **Liang–Barsky** – parametric line P = P1 + u(P2 − P1), 0 ≤ u ≤ 1; clip with inequalities pu ≤ q for each edge – faster, fewer intersections.
- **Sutherland–Hodgman polygon clipping** – clip the polygon against one window edge at a time (left, right, bottom, top); four cases for each edge of the polygon: in→in, in→out, out→out, out→in.

16. Hidden Line / Hidden Surface Removal

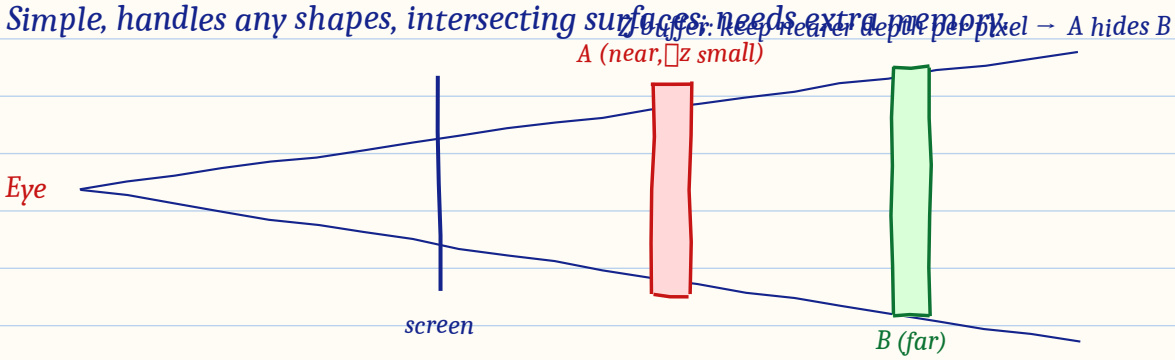
- When a 3D object is shown on a 2D screen, edges / faces that are behind other faces must not be shown, otherwise the picture is ambiguous. The process is **hidden line (HLR) / hidden surface removal (HSR)**.
- Two classes of algorithms:
  - **Object-space methods** – compare objects / faces with each other (accurate, device independent) e.g. back-face removal, depth sorting, Roberts algorithm.
  - **Image-space methods** – decide visibility pixel by pixel on the screen, e.g. Z-buffer, scan-line, Warnock's area subdivision.

Back-face removal (culling)

- A polygon is **back-facing** if its outward normal N makes an angle > 90° with the viewing direction V, i.e. **$N \cdot V > 0$** (when V points from eye to object). Such faces are discarded (removes about half the faces of a closed solid).
- Works only for convex solids; for concave objects other tests are also needed.

Z-buffer (depth-buffer) algorithm

- Keep two arrays: **frame buffer** (colour) and **Z-buffer** (depth of each pixel). Initialise depth = far, colour = background.
- For each polygon and each pixel it covers, compute depth z; if z is nearer than the stored value → update depth and colour; else ignore.
- Simple, handles any shapes, intersecting surfaces; needs extra memory.



Painter's (depth-sort) algorithm

- Sort polygons by depth from farthest to nearest and paint them in that order – nearer faces overwrite farther ones.

Scan-line and Warnock algorithms

- **Scan-line**: process one horizontal line at a time; for each span find the nearest polygon.
- **Warnock**: recursive subdivision of the window until in each area the visible face is

obvious (one polygon, none, or the nearest).

Hidden line methods

- Wire-frame display: edges tested against faces; **Roberts algorithm** – for convex polyhedra; removes back faces, then edges hidden by other faces.

17. Colour and Shading

Colour models

- **RGB** (additive – used on screens: red, green, blue; each 0 to 255).
- **CMY / CMYK** (subtractive – used in printing: cyan, magenta, yellow, black).
- **HSV / HLS** – hue, saturation, value – intuitive for user selection.
- Colour depth: 8-bit = 256 colours, 24-bit "true colour" = 16.7 million; colour look-up table (LUT) reduces memory.

Shading / illumination models

- **Shading** = calculation of light intensity on each point of a surface to give a 3D realistic look.
- Light components: **ambient** (uniform background light), **diffuse** (Lambert's cosine law, $I = Id kd \cos\theta$, rough surfaces), **specular** (shiny highlight, Phong term $ks \cos^n \alpha$).

$$I = Ia \cdot ka + Id \cdot kd \cdot (N \cdot L) + Is \cdot ks \cdot (R \cdot V)^n$$

Shading methods for polygon meshes

Method	Idea	Remark
Flat (constant)	one intensity for the whole polygon	fast, faceted look
Gouraud	intensity at vertices, linear interpolation across polygon	smooth; may miss highlights
Phong	interpolate normals, compute intensity at each pixel	best quality, slow

- **Ray tracing** – follows rays from eye through each pixel to the scene; gives reflection, refraction, shadows (photo-realistic but slow). **Rendering** = shading + shadows + texture + transparency.

18. Quick Revision – Important Points

CAD → design side | CAM → manufacturing side | CIM → whole enterprise on one database.

- *Product cycle: design cycle + manufacturing cycle; CAD/CAM links them.*
- *CAD functions: geometric modelling, engineering analysis, design review, drafting.*
- *CAM areas: computer monitoring & control and manufacturing support.*
- *Data structures: array, linked list, tree, network, relational; DBMS models: hierarchical, network, relational, object-oriented.*
- *Coordinate systems: MCS, WCS, SCS; Cartesian, cylindrical, spherical.*
- *Transformations use homogeneous coordinates; concatenation order matters (TR ≠ RT).*
- *Clipping: Cohen–Sutherland (outcodes), Liang–Barsky, Sutherland–Hodgman.*
- *HSR: back-face, Z-buffer, painter's, scan-line, Warnock. Shading: flat, Gouraud, Phong.*

Likely exam questions

- *Explain the product cycle and the role of CAD / CAM / CIM in it.*
- *Differentiate between CAD, CAM and CIM; list benefits of CIM.*
- *Discuss the criteria for evaluating a CAD/CAM system.*
- *What is a DBMS? Explain the different database models used in CAD.*
- *Derive the 2D rotation matrix; explain rotation about an arbitrary point.*
- *Explain the Cohen–Sutherland line clipping algorithm with a neat sketch.*
- *Explain any two hidden surface removal algorithms. Compare Gouraud and Phong shading.*

UNIT – II : Geometric Modeling

1. Introduction

- **Geometric modeling** = mathematical description of the shape (geometry + topology) of an object so that a computer can store, display, analyse and manufacture it.
- The model is the heart of CAD: the same model feeds drafting, analysis (FEA), NC tool-path generation, mass-property calculation and rendering.
- Three types of CAD models:
  - **Wire-frame model** – edges / curves only (skeleton).
  - **Surface model** – skin of the object defined by surface patches.
  - **Solid model** – complete volume with inside / outside information.
- Uses of geometric modeling: design, mass properties, interference check, finite-element mesh, machining simulation, animation.

2. Representation Techniques

- **Explicit (non-parametric)**: $y = f(x)$, $z = g(x)$ – one variable gives the others.
  - Example: straight line $y = mx + c$; circle $y = \pm\sqrt{(r^2 - x^2)}$.
- **Implicit (non-parametric)**: $f(x, y) = 0$ (or $f(x, y, z) = 0$).
  - Example: circle $x^2 + y^2 - r^2 = 0$; sphere $x^2 + y^2 + z^2 = r^2$.
- **Parametric**: every coordinate is a function of a parameter(s): $x = x(u)$, $y = y(u)$, $z = z(u)$ with u from 0 to 1 (curve) or $P(u, v)$ (surface).
  - Circle: $x = r \cos u$, $y = r \sin u$; line: $P(u) = P1 + u(P2 - P1)$.

Parametric vs non-parametric

Non-parametric	Parametric
Depends on the axes; slope may be infinite (vertical line)	Independent of axes; tangent vector always finite
Multi-valued curves (circle) need several equations	Closed / multi-valued curves easy, one equation set
Bounded segments need extra limits	Bounds are built in ($0 \le u \le 1$)
Point on curve found by solving equation	Points generated directly by varying u
Difficult transformations	Simple transformations of control points

CAD systems prefer the **parametric** form: easy to compute, plot, transform, and to join curves.

3. Wire-frame Modeling

- Object represented by **points, lines, circles, arcs and curves** (edges); no surface or volume information.
- Types: 2D, 2½D (extruded) and 3D wire frames.
- Data stored: a **vertex table** (coordinates) and an **edge table** (pair of vertices).

Advantages

- Simple, needs little memory and computing time; fast to draw and edit.
- Gives all views easily; good for rough design, layouts and as the base for surface / solid models.

Limitations

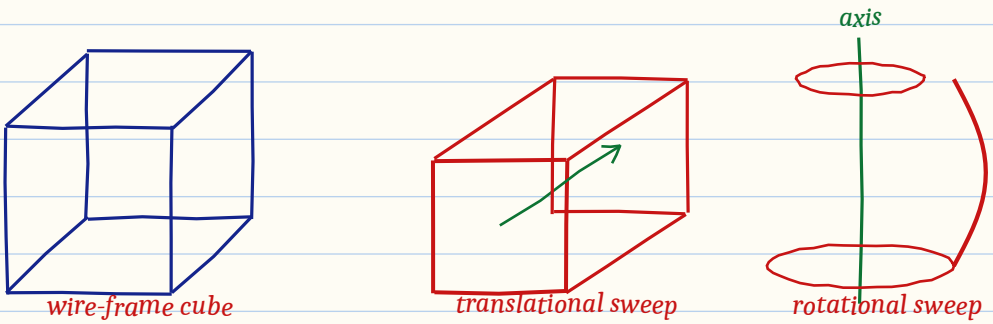
- **Ambiguity** – the same wire-frame can be interpreted as several solids (e.g. Necker cube).
- No hidden-line removal automatically; cannot calculate volume / mass; no section views, no shading; no information for NC machining of faces; interference checking not possible.

4. Construction Methods (creating wire-frame / basic geometry)

- **Points**: by coordinates, by intersection of lines, at the end / mid-point of an entity, on a curve at distance.
- **Lines**: two points; point + angle + length; parallel / perpendicular / tangent to another entity; horizontal / vertical.
- **Circles & arcs**: centre + radius; three points; two points + radius; tangent to two

entities (TTR); fillet / chamfer.

- **Ellipse, parabola, hyperbola** (conics) by centre / foci and axes.
- Editing: trim, extend, offset, mirror, copy, array, scale, rotate.
- **Sweeping**: translational sweep (extrude a profile along a line) and rotational sweep (revolve around an axis).



5. Synthetic Curves (free-form curves)

- Analytic curves (line, circle, conic) cannot describe free-form shapes such as car bodies, turbine blades, ship hulls → **synthetic curves** are built from **data points / control points** by mathematical formulation.
- Two approaches:
  - **Interpolation** – curve passes exactly through the given points (Hermite cubic, cubic spline).
  - **Approximation** – curve is shaped by control points but need not pass through them (Bezier, B-spline, NURBS).
- Desirable properties: smoothness (continuity), local control, ease of modification, low degree (cubic gives best compromise), invariance under transformation.
- **Continuity**: C0 – curves just meet; C1 – same tangent direction (slope); C2 – same curvature. (G0, G1, G2 = geometric continuity).

5.1 Hermite cubic spline

- Defined by two end points P1, P2 and two end tangent vectors P1′, P2′.

$$P(u) = F1\,P1 + F2\,P2 + F3\,P1' + F4\,P2', \quad 0 \le u \le 1$$

- Blending functions: $F1 = 2u^3 - 3u^2 + 1$, $F2 = -2u^3 + 3u^2$, $F3 = u^3 - 2u^2 + u$, $F4 = u^3 - u^2$.
- Matrix form: $P(u) = [u^3\ u^2\ u\ 1]\,[M_H]\,[P1\ P2\ P1'\ P2']^T$

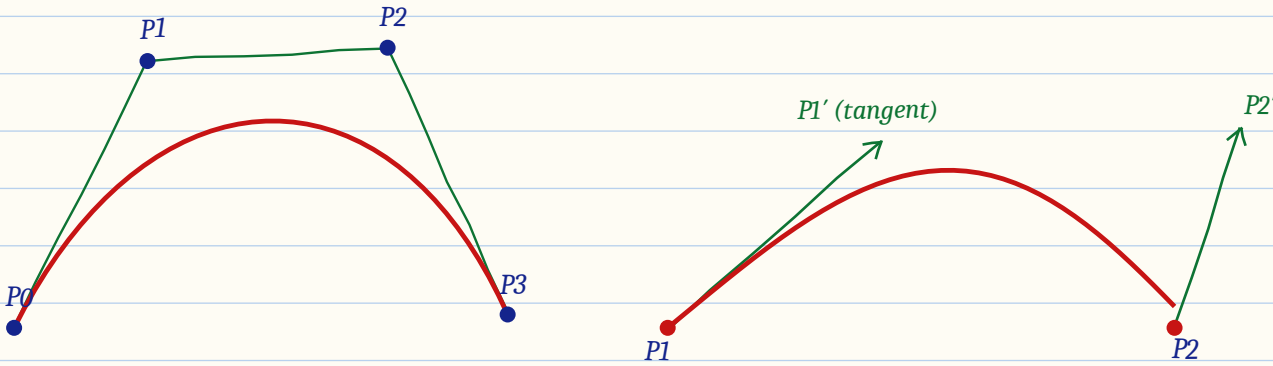
$$[M_H] = \begin{bmatrix} 2 & -2 & 1 & 1 \\ -3 & 3 & -2 & -1 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

- Check: $u = 0 \rightarrow P1$; $u = 1 \rightarrow P2$. Tangent magnitude controls the "fullness" of the curve; two Hermite segments joined with equal tangent → C1 continuity.
- Drawback: user must give tangents – not intuitive.

5.2 Bezier curve

- Defined by **n + 1 control points** P0 … Pn (control polygon); degree = n.

$$\mathbf{P(u) = \Sigma\ Pi \cdot B(i,n)(u)\ ,\quad B(i,n)(u) = C(n,i)\ u^i\ (1-u)^{n-i}}$$



Bezier curve (cubic) Hermite cubic

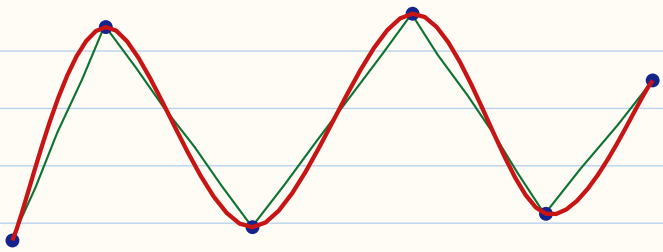
- Cubic Bezier (n = 3): $P(u) = (1-u)^3P0 + 3u(1-u)^2P1 + 3u^2(1-u)P2 + u^3P3$.
- **Properties**:
  - Curve passes through the first and last control points only (P0 and Pn).
  - Tangent at P0 is along P0P1; tangent at Pn is along Pn−1Pn.
  - Curve lies inside the **convex hull** of the control polygon; blending functions are all positive and add to 1.
  - **Variation diminishing** – no wiggles beyond the polygon; symmetric about u = 0.5.
  - Degree = number of control points − 1, so many points give a high-degree curve; **no local control** (moving one point changes the whole curve).
  - Joining segments: C1 continuity if P3 of first, P0 of second and neighbours are collinear.

5.3 B-spline curve

- Also defined by control points but made of **piecewise polynomial segments** of degree p chosen independently of the number of control points.

$$P(u) = \Sigma\, Pi \cdot N(i,k)(u), \quad i = 0 \ldots n \quad (k = order = degree + 1)$$

- Basis functions N(i,k) are found from a **knot vector** (non-decreasing set of parameter values) by the Cox–de Boor recursion formula.
- **Properties**: local control (a control point affects only k segments), degree independent of number of points, built-in continuity C(k−2), convex-hull property, curve does not pass through control points (unless the knots are repeated – clamped end conditions).
- Types: uniform (equal knot spacing), non-uniform, open (clamped) – the open B-spline passes through first and last points.



moving P3 changes only nearby segments

B-spline: local control (smooth, approximates polygon)

5.4 NURBS

- **Non-Uniform Rational B-Splines**: B-spline with a **weight** wi on every control point and a non-uniform knot vector.

$$P(u) = \Sigma\, wi\, Pi\, N(i,k)(u) \;/\; \Sigma\, wi\, N(i,k)(u)$$

- Advantages: can exactly represent **conics (circle, ellipse, parabola)** and free-form shapes in one form; weights give extra shape control (larger weight pulls the curve towards the point); invariant under projective transformation. Standard in CAD (IGES, STEP).

Comparison

Curve	Passes through points	Local control	Remark
Hermite	all end points	no	needs tangents
Bezier	first & last	no	simple, degree = n
B-spline	generally not	yes	degree independent of n
NURBS	generally not	yes	exact conics + free-form

6. Surface Modeling

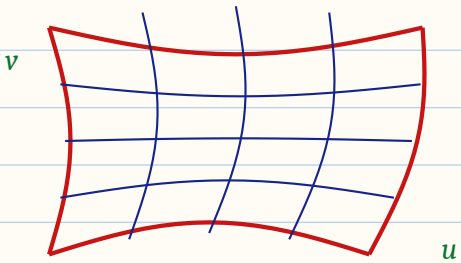
- Surface model describes the **outer skin** of an object using surface patches; it has **no volume (inside) information**.
- Used where the shape is complex or aesthetic: automobile bodies, aircraft fuselage, ship hulls, turbine blades, dies & moulds, consumer products.

Advantages and limitations

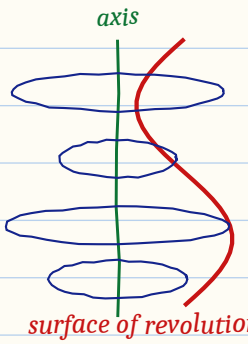
- Unambiguous shape, hidden-line / surface removal, shading, 5-axis NC tool paths, intersection of surfaces; better than wire-frame.
- Limitations: no mass properties directly; solid information missing, difficult to ensure a closed watertight body; more data and skill needed.

7. Synthetic Surfaces

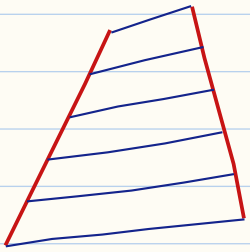
- **Analytic surfaces**: plane, cylinder, cone, sphere, torus (equations known).
- **Synthetic (free-form) surfaces**: created from points / curves – the surface analogue of curves **P(u,v)**, with two parameters u and v ($0 \le u, v \le 1$).



Surface patch P(u,v)



surface of revolution


ruled surface

7.1 Plane surface

- Defined by three non-collinear points or a point and a normal: $P(u, v) = P0 + u(P1 - P0) + v(P2 - P0)$.

7.2 Ruled (lofted) surface

- Generated by joining corresponding points of two boundary curves P(u) and Q(u) with straight lines (rulings):

$$S(u, v) = (1 - v)\, P(u) + v\, Q(u)$$

- Examples: cone, cylinder, hyperboloid; used for turbine blades.

7.3 Surface of revolution

- A plane curve (**generatrix**) is revolved about an axis through angle θ: $x = f(u)\cos\theta$, $y = f(u)\sin\theta$, $z = g(u)$.
- Examples: vases, pulleys, bottles, domes, lathe-turned parts.

7.4 Tabulated cylinder

- A plane curve is translated along a straight line (direction vector) – it is the surface of a translational sweep (extrusion).

7.5 Hermite bicubic surface

- 16 vectors: 4 corner points, 8 tangent vectors (u and v at corners) and 4 twist vectors.

$$P(u, v) = \Sigma\, \Sigma\ aij\, u^i\, v^j \quad (i, j = 0 \ldots 3)$$

- Patch can be joined with C1 continuity to neighbours.

7.6 Bezier surface

- Uses a **rectangular net** of $(m + 1) \times (n + 1)$ control points.

$$P(u,v) = \Sigma i\, \Sigma j\, Pij\, Bi,m(u)\, Bj,n(v)$$

- Properties are the extension of the Bezier curve: corner points are on the surface, lies inside the convex hull of the net, edge curves are Bezier curves.

7.7 B-spline and NURBS surfaces

- Tensor product of B-spline basis functions in u and v with knot vectors U and V → local shape control; NURBS adds weights → exact spheres, cylinders, tori.

7.8 Coons patch

- Surface built by blending four boundary curves P(u,0), P(u,1), P(0,v), P(1,v) – used to fill a closed boundary (**surface blending**).

Other surface operations

- **Fillet / blend** surfaces, **offset** surface, trimming, **sweep** and **loft** (skinning through several sections), **extension**, intersection.

8. Solid Modeling

- **Solid model** = complete, unambiguous, informationally complete representation of the volume: it knows which points are **inside, outside or on the boundary** of the object.
- Basic requirements (valid solid): **rigid** (shape independent of position), **homogeneous (3D)** – no dangling edges / faces, **finite** (bounded), closed under rigid motion and Boolean operations, **finitely describable**, boundary determinism.
- Advantages: unambiguous, mass properties (volume, centroid, inertia), interference check, section views, FEA mesh, NC verification, assembly modeling.
- Disadvantages: needs more memory and computing power; free-form shapes more difficult.

Solid representation schemes

- **Boundary representation (B-rep)**, **Constructive solid geometry (CSG)**, **Sweep representation**, **Spatial enumeration (voxels / octree)**, **Cell decomposition**, **Primitive instancing**, **Feature-based (parametric history) modeling**.

9. Fundamentals of Solids – Primitives and Boolean Operations

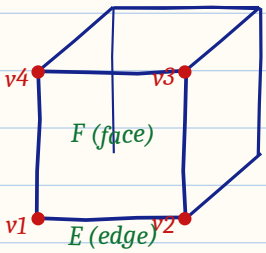
- **Primitives** (basic solids): block / box, cylinder, cone, sphere, torus, wedge, pyramid.
- **Boolean (regularised set) operations**:
  - **Union (∪)** – joins A and B into one solid.
  - **Intersection (∩)** – common volume of A and B.
  - **Difference (–)** – removes B from A (hole, slot, pocket).
- Regularisation ensures the result is again a valid solid (no dangling faces / edges).
- Other: **sweeping** (extrude, revolve, helical sweep), **blending / filleting**, **chamfering**, **shelling**, **draft**.
- **Euler's formula** for a valid simple polyhedron:

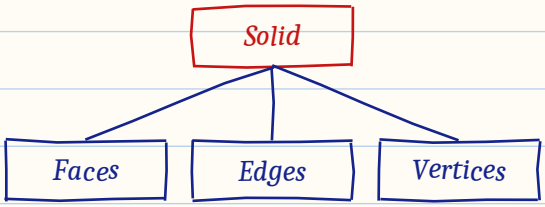
$$V - E + F = 2 \quad \text{(cube: } 8 - 12 + 6 = 2\text{)}$$

- For solids with holes (genus g): $V - E + F - R = 2(S - g)$ (R = inner loops, S = shells).

10. Boundary Representation (B-rep)

- Solid is described by its **boundary** – set of **faces** bounded by **edges** that end at **vertices**; the surface separates inside from outside.
- Data stored in two parts:
  - **Geometry** – equations of surfaces, curves and coordinates of vertices.
  - **Topology** – how faces, edges, vertices are connected (adjacency), and the orientation (outward normal) of each face.





Topology: F, E, V relations

Euler: V – E + F = 2

- Typical structure (winged-edge): Solid → Faces → Loops → Edges (with two adjacent faces and two end vertices) → Vertices.

Advantages

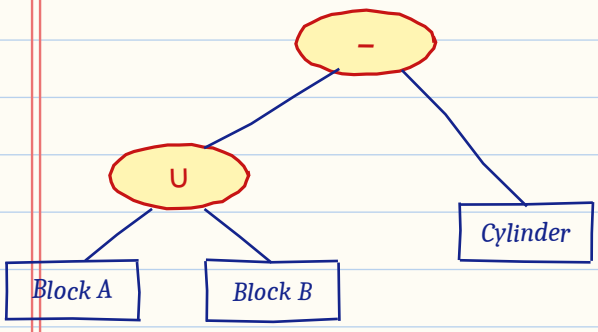
- Fast display and hidden-line removal (faces are explicit); good for free-form surfaces; easy conversion to wire-frame; direct link to surface modeling; good for NC.

Disadvantages

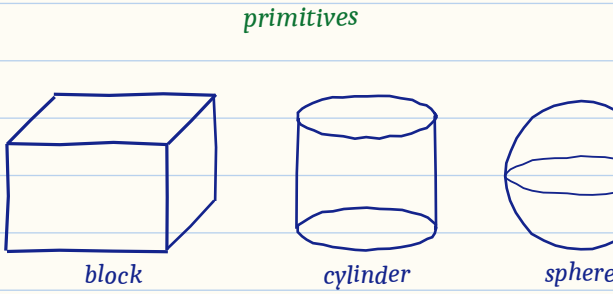
- Large storage; complex data structure; validity must be maintained (Euler operators); cannot directly show the design history; Boolean operations are complex.

11. Constructive Solid Geometry (CSG)

- Solid is built by combining **primitive solids** with **Boolean operations** and rigid motions; stored as a **binary tree**: leaves = primitives (with size & position), nodes = operations (∪, ∩, −) or transformations.



CSG tree: (A ∪ B) – Cylinder



- Evaluation of the tree gives the final object; changing a primitive's size (e.g. hole diameter) is easy (parametric).

Advantages

- Compact data, always valid solids, simple to create, easy modification, records design history, easy to compute volume & mass.

Disadvantages

- Only the tree is stored → boundary (faces / edges) must be **evaluated** for display (slow); limited to Boolean operations, difficult to create sculptured surfaces; not unique (the same solid can have different trees).

B-rep vs CSG

B-rep	CSG
Stores faces, edges, vertices	Stores primitives + Boolean tree
Large data; fast display	Compact; slow display (needs evaluation)
Easy free-form shapes	Limited to primitives
Needs validity checks	Always valid
Boundary evaluated already	Boundary must be computed

- **Hybrid modelers** (CSG + B-rep) are used in practice (CATIA, NX, SolidWorks): the CSG tree is the master and B-rep is kept for display.

12. Other Solid Schemes (brief)

- **Sweep representation**: a 2D profile moved along a path (translational, rotational, general sweep). Good for extruded / turned parts.
- **Spatial enumeration**: space divided into small cubes (voxels); **octree** divides recursively – used in medical imaging and 3D printing slicing.
- **Cell decomposition**: solid built from simple cells glued together – FEM meshes.
- **Feature-based modeling**: parametric features (hole, slot, rib, fillet) with dimensions

and constraints; changing a dimension regenerates the model.

12A. Comparison of the Three Modeling Methods

Feature	Wire-frame	Surface	Solid
Information stored	vertices & edges	edges + surface patches	faces, edges, volume, inside / outside
Ambiguity	high	low	none
Hidden line removal	not possible / manual	possible	automatic
Mass properties	no	no (approx.)	yes – volume, centre of gravity, inertia
Memory & time	least	medium	most
NC / FEA use	poor	good (surface machining)	excellent
Typical use	layouts, skeleton, mechanisms	car body, blades, dies	machine parts, assemblies

12B. Worked Examples

Example 1 – Bezier curve point

- Control points P0(0,0), P1(2,4), P2(5,4), P3(7,0); find the point at $u = 0.5$.
- Blending values ($n = 3$, $u = 0.5$): $(1-u)^3 = 0.125$, $3u(1-u)^2 = 0.375$, $3u^2(1-u) = 0.375$, $u^3 = 0.125$.
- $x = 0.125(0) + 0.375(2) + 0.375(5) + 0.125(7) = 0.75 + 1.875 + 0.875 =$ **3.5**
- $y = 0.125(0) + 0.375(4) + 0.375(4) + 0.125(0) = 1.5 + 1.5 =$ **3.0**

P(0.5) = (3.5, 3.0) – lies inside the convex hull, as the property says.

Example 2 – Hermite curve

- P1(0,0), P2(4,0), tangents P1′ = (0,4), P2′ = (0,–4). At $u = 0.5$: F1 = 0.5, F2 = 0.5, F3 = 0.125, F4 = –0.125.
- $x = 0.5(0) + 0.5(4) + 0.125(0) - 0.125(0) = 2$.
- $y = 0 + 0 + 0.125(4) - 0.125(-4) = 0.5 + 0.5 = 1$.

P(0.5) = (2, 1) → the curve bulges upward between the two end points because of the tangents.

Example 3 – Euler's formula check

- Cube: $V = 8, E = 12, F = 6 \rightarrow 8 - 12 + 6 = 2$ ✔.
- Triangular prism: $V = 6, E = 9, F = 5 \rightarrow 6 - 9 + 5 = 2$ ✔.
- Cube with a through-hole (genus 1): $V = 16, E = 24, F = 10 \rightarrow V - E + F = 2 - 2g = 0$ ✔ (so the simple formula is changed for holes).

Example 4 – Parametric equations

- Line from A(2,1) to B(6,5): $x = 2 + 4u, y = 1 + 4u, 0 \le u \le 1$.
- Circle of radius 5 at origin: $x = 5 \cos u, y = 5 \sin u, 0 \le u \le 2\pi$.
- Ellipse a, b: $x = a \cos u, y = b \sin u$. Helix: $x = r \cos u, y = r \sin u, z = pu/2\pi$.
- Parabola: $x = u, y = a\,u^2$ (non-parametric $y = a\,x^2$).

12C. Features and Parametric Modeling (modern practice)

- **Feature** = an engineering shape with meaning (hole, slot, pocket, boss, rib, fillet, chamfer, thread) – created with dimensions, not with raw primitives.
- **Parametric (history-based) modeling**: the model stores the sequence of operations (feature tree) and dimensions; changing a dimension regenerates the part.
- **Constraints**: geometric (parallel, tangent, concentric, perpendicular) and dimensional (length, angle, diameter) – make a sketch fully defined.
- **Variational vs parametric**: variational solves all constraints simultaneously (no fixed order); parametric solves in order of creation.
- **Assembly modeling**: mates / constraints between parts (coincident, concentric, distance), interference check, exploded views, bill of materials.
- **Direct modeling**: push / pull faces without the history tree – quick edits of imported geometry.
- **Data exchange**: IGES (curves and surfaces), STEP (full product data), STL (triangle mesh for 3D printing), DXF (2D drawing).

13. Quick Revision

Representation: explicit $y = f(x)$, implicit $f(x,y)=0$, parametric $P(u)$.

- Curves: Hermite (end points + tangents), Bezier (control polygon, no local control), B-spline (local control), NURBS (weights, exact conics).
- Continuity: C0 position, C1 tangent, C2 curvature.
- Surfaces: plane, ruled, revolution, tabulated cylinder, Hermite bicubic, Bezier,

B-spline, NURBS, Coons.

- Wire-frame < Surface < Solid (in information content). Wire-frame is ambiguous; solid is unambiguous.
- Solid schemes: B-rep (faces, edges, vertices; Euler V−E+F=2) and CSG (primitives + Boolean tree).

Likely exam questions

- Compare parametric and non-parametric representation of curves.
- Derive the Hermite cubic curve equation with blending functions.
- Explain Bezier curve, its properties and advantages over B-spline.
- Write short notes on wire-frame modeling: advantages and limitations.
- Explain the various synthetic surfaces with sketches.
- Differentiate between B-rep and CSG solid modeling schemes with neat figures.
- Explain Boolean operations and Euler's formula with examples.

UNIT – III : Numerical Control & CNC Part Programming

1. Numerical Control (NC)

- **NC** = a form of programmable automation in which the operations of a machine tool are controlled by **numbers, letters and symbols** (coded instructions) that are fed to the machine. (EIA definition)
- Invented at MIT (1952, J. Parsons) for milling aircraft parts; now every machine tool is NC / CNC.
- The machine follows the program automatically: tool position, speed, feed, coolant, tool change.

Why NC?

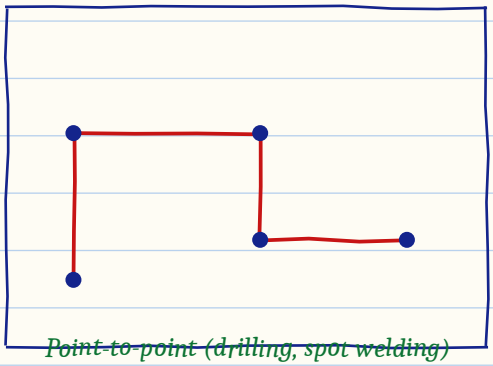
- High accuracy and repeatability, complex shapes, less setup time, small-batch (job-shop) flexibility, lower scrap, reduced operator skill & fatigue, less jigs and fixtures, easy design changes by editing the program.

Limitations

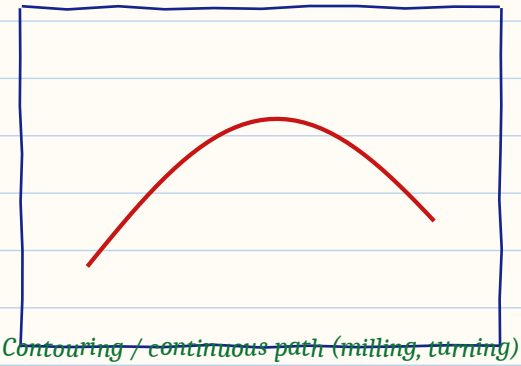
- High initial cost, skilled programmers and maintenance, not economical for very large batches (special-purpose machines are better), computer / electronic problems.

2. NC Modes (types of control / motion control)

- **Point-to-point (PTP) / positioning control**: tool moves to a position at rapid speed; machining is done only after reaching it (drilling, boring, tapping, spot welding, punching). Path between points is unimportant.
- **Straight-cut (line) control**: machining along a straight line parallel to an axis at controlled feed (simple milling, turning, facing).
- **Contouring (continuous path) control**: two or more axes move simultaneously with a coordinated speed ratio to cut any path – angled lines, arcs, curves (milling, lathe, profile cutting). Needs an **interpolator** (linear, circular, parabolic, cubic).



Point-to-point (drilling, spot welding)

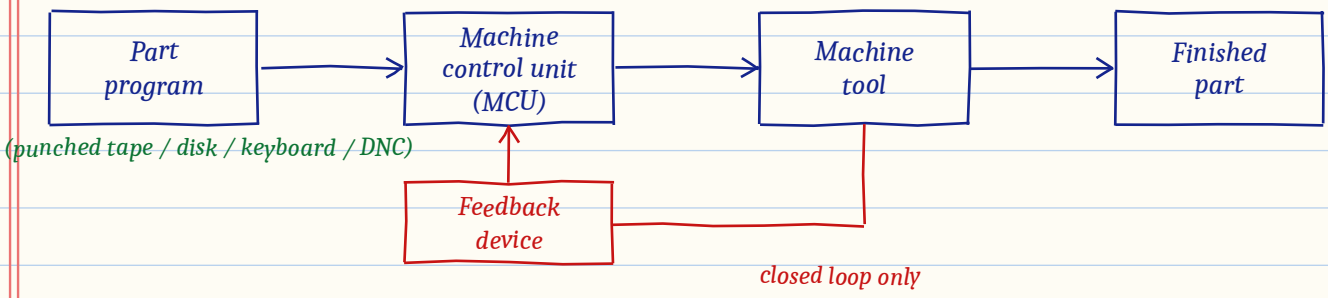


Contouring / continuous path (milling, turning)

- Programming modes: **absolute** (all dimensions measured from a fixed zero: G90) and **incremental** (each move measured from the previous point: G91).

3. Elements of an NC System

- Three basic elements:
  - **Program of instructions (part program)** – step-by-step commands in coded form (tape, disk, memory).
  - **Machine control unit (MCU)** – electronics that reads the program, decodes it and sends signals to drive the machine (data processing unit + control loops unit).
  - **Machine tool (processing equipment)** – the machine that does the cutting (with slides, spindle, table, drives, feedback devices).



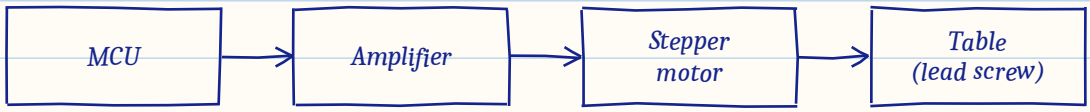
- **Data processing unit (DPU)**: tape reader, data buffer, decoding, interpolator, sends signals to the control loops unit.
- **Control loops unit (CLU)**: servo-drives for each axis, receives feedback, signals for spindle speed, coolant, tool change.

Open-loop and closed-loop systems

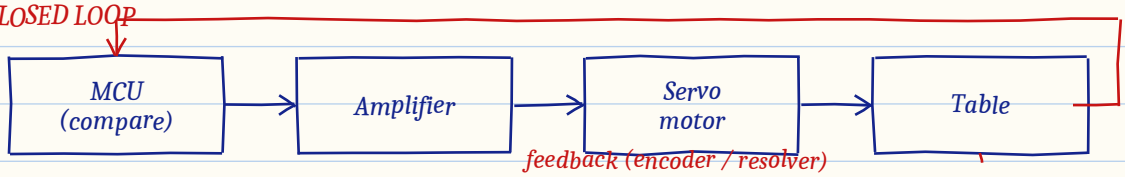
- **Open loop**: uses **stepper motor**, no feedback of actual position; simple, cheap, used for light loads; accuracy depends on the motor.
- **Closed loop**: **servo motor** (DC / AC) with **feedback** (encoder, resolver,

tachogenerator, LVDT); the actual position is compared with the commanded position and the error is corrected. Accurate but costly.

OPEN LOOP



CLOSED LOOP



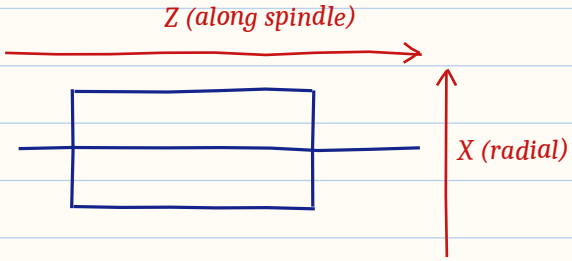
- Feedback devices: **encoder** (optical, incremental or absolute), **resolver** (analog, rotary transformer), **linear scale / inductosyn**, **tachogenerator** (speed).

4. NC Machine Tools and their Structure

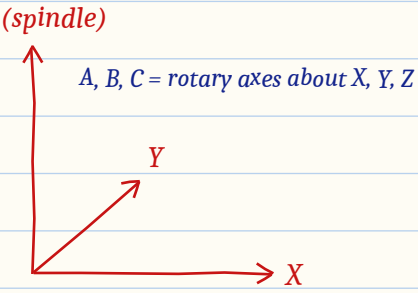
- Typical structure: **Bed / base** (cast iron or polymer concrete, rigid, damping) → **Column** → **Slides / guideways** → **Table** → **Spindle & headstock** → **Drives** → **Control panel** (MCU) → **Tool changer** and **coolant** system.
- Special design features of NC machines:
  - **High static and dynamic rigidity**, low thermal distortion (symmetrical design, cooling of spindle).
  - **Low-friction slideways**: hydrostatic ways or **linear motion (LM) guideways** with rolling elements; PTFE-coated (turcite) slideways to avoid stick-slip.
  - **Recirculating ball screw and nut** for table drive (efficiency about 90%, zero backlash with preloaded double nut).
  - **Backlash elimination**, quick-response servo drives, wide spindle speed range (variable speed AC / DC drive).
  - **Automatic tool changer (ATC)**, tool magazine, chip conveyors, enclosed safe guarding, in-process probing.

Axes of motion (ISO / EIA convention)

- **Z-axis** = along the spindle axis (positive direction = away from workpiece); **X-axis** = main horizontal axis; **Y-axis** by right-hand Cartesian rule; **A, B, C** = rotations about X, Y, Z.



CNC lathe (2 axes: X, Z)



Vertical milling (X, Y, Z)

NC lathe (turning centre)

- Axes X (cross slide) and Z (carriage); turret with 8 to 12 tools; chuck, tailstock; live tooling and C-axis in turn-mill centres.

NC milling machine

- Vertical or horizontal spindle; table moves in X, Y, knee / head in Z; 3-axis, 4-axis (rotary table) and 5-axis machines.

5. Machining Centre – types and features

- **Machining centre (MC)** = NC machine that can do **several machining operations** (milling, drilling, boring, tapping) on a workpiece in **one setup**, with **automatic tool changing** from a tool magazine and often an automatic pallet changer.

Types

- **Vertical machining centre (VMC)** – vertical spindle; for flat parts, dies and moulds; 3 axes.
- **Horizontal machining centre (HMC)** – horizontal spindle; for box-type parts; chips fall away; pallet changer, rotary table (B-axis).
- **Universal machining centre (UMC)** – 5-axis; tilting spindle or table; complex parts in one setup.
- **Turning centre / turn-mill centre** – lathe with live tools; **Gantry-type** machining centres for very large parts.

Features

- Automatic tool changer (ATC): 20 to 100+ tools, tool change in 2 to 8 seconds; tool identification by pocket number (random access) or tool code.
- **Automatic pallet changer (APC)** – load / unload while machining another pallet (reduces idle time).

- Rotary table indexing, probing for work-offset and tool-length measurement, high spindle speed (6000 to 30000 rpm), through-spindle coolant, chip conveyor.
- High productivity, accuracy, flexibility; basis of **FMS cells**.

6. Controls in NC / CNC

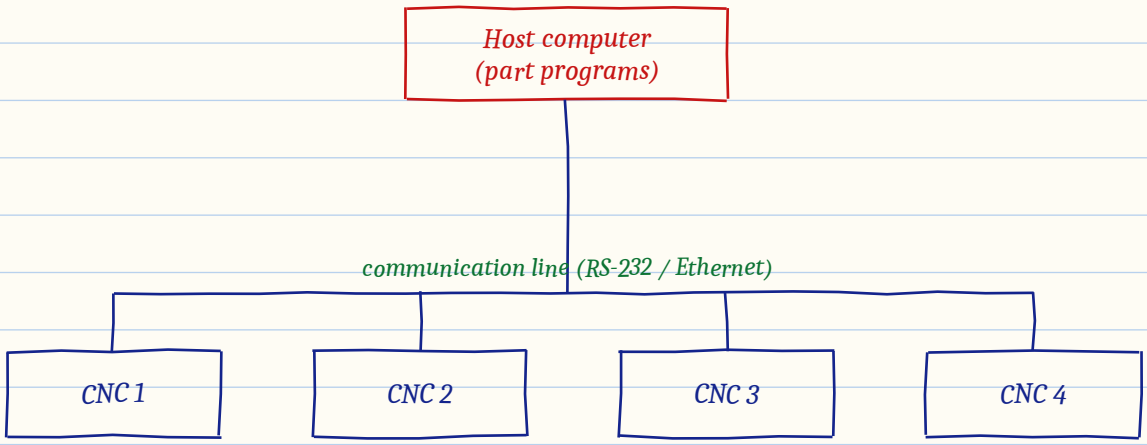
- Control types by hardware: **hard-wired NC** (fixed logic, early) → **soft-wired CNC** (microcomputer inside MCU; program kept in memory; can be edited at the machine).

CNC – Computer Numerical Control

- **CNC** = an NC system in which a dedicated **microcomputer** performs the basic NC functions by stored software (**executive program**).
- Features: program stored in RAM / memory, **editing at the machine**, graphic display and simulation, subprograms and macros (parametric programming), canned cycles, cutter-radius and tool-length compensation, diagnostics, communication ports, adaptive control possibility.
- Advantages over NC: more flexible, less hardware, easy upgrade by software, higher reliability, tape reader not required, program storage and reuse, accuracy through compensation of errors.

7. DNC – Direct (Distributed) Numerical Control

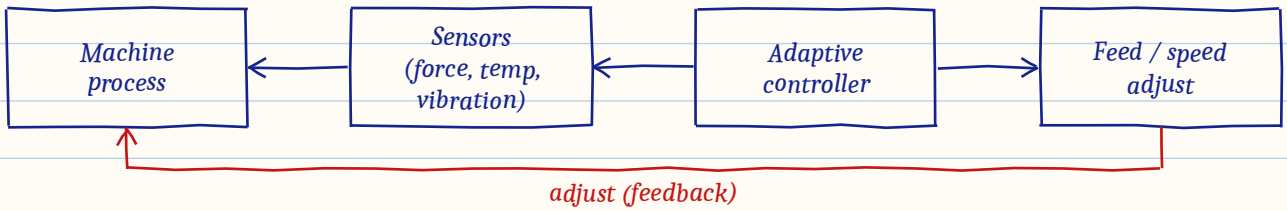
- **DNC** = use of a **central (host) computer** to control a number of NC / CNC machines by direct connection; the part programs are stored in the host and sent to the machines on demand (block by block or whole program).
- Two types:
  - **Direct DNC** – host computer is directly connected and controls the machines (no tape reader); if the computer fails all machines stop.
  - **Distributed DNC** – host sends programs to individual CNC machines with their own memory through a **network (LAN)**; machines continue even if the host is down.



- Components: central computer (programs), bulk memory, communication lines, machine tools (with MCU), software for program storage, download and tracking of production data.
- Advantages: **elimination of tape**, large program storage, easy program editing & transfer, production monitoring & data collection, scheduling and basis for FMS and CIM.

8. Adaptive Control Machining Systems

- **Adaptive control (AC)**: the machining conditions (feed, speed) are **automatically adjusted in real time** to the actual conditions of the process measured by sensors – force, torque, temperature, vibration, tool wear, power.
- Reason: conventional NC uses fixed feed / speed set conservatively; actual cutting varies due to hardness variation, tool wear, depth of cut, etc.



Types of adaptive control

- **ACC – Adaptive Control with Constraints**: a measured variable (cutting force, torque or power) is kept **below a limit** by adjusting the feed (or speed). Protects tool and machine; most common in practice.
- **ACO – Adaptive Control with Optimisation**: an **index of performance** (e.g. minimum cost per part or maximum production rate) is calculated on line and the feed /

speed are adjusted to optimise it (needs tool-wear / cost models – complex).

- **GAC – Geometric Adaptive Control**: compensates for geometric deviations (tool deflection, thermal growth) to maintain **dimensional accuracy and surface finish**.

Benefits

- Higher metal-removal rate, longer tool life, protects the machine & tool from overload, less operator involvement, better surface finish. Limits: costly sensors, sensor reliability in a harsh environment.

CNC Part Programming

9. Fundamentals of Part Programming

- **Part program** = set of instructions in a coded form that tells the machine what to do to make the part.
- Steps: study the drawing → select process, machine, tools, cutting data → fix the **work datum (zero point)** and clamping → decide tool path & sequence → write the program → enter on machine → **dry run / simulation** → prove out → production.
- Methods: **manual part programming** (G & M codes) and **computer-assisted part programming** (APT, Mastercam; geometry is defined, then the computer calculates the tool path and a post-processor converts it into machine code).
- Coordinate system: Cartesian right-hand rule. **Machine zero** (fixed by the maker) and **work zero / program zero** (set by the user, G54 to G59). **Reference point** is used for homing.
- Data to be decided: spindle speed $N = 1000\,V / (\pi D)$ rpm, feed (mm/min = $f \times N$), depth of cut, tool numbers & offsets, coolant.

10. NC Word and Block (Program Format)

- **Character** (letter, digit, symbol) → **word** (letter + number) → **block** (line = one operation) → **program**.
- **NC word** = address letter + numerical value. It gives one type of instruction.

Address	Meaning	Example
N	sequence (block) number	N010
G	preparatory function – mode of motion	G01 linear interpolation
X Y Z	coordinate positions	X50.0 Y25.0
I J K	arc centre (incremental) / arc parameters	I10 J0
F	feed rate (mm/min or mm/rev)	F120
S	spindle speed (rpm)	S1200
T	tool number	T03
M	miscellaneous (auxiliary) function	M03 spindle on
D / H	tool radius / length offset number	D01 H01

- **Block format**: *word address format* is standard (words may be omitted if

unchanged – modal). Example block:

N040 G01 X60.0 Y30.0 F150 S800 M03 ;

- **Modal** codes remain active until changed; **non-modal** act only in the block where they are written.
- Program starts with % / O-number, ends with M30 (end & rewind) or M02.

Important G codes (preparatory functions)

Code	Function	Code	Function
G00	rapid positioning	G01	linear interpolation (feed)
G02	circular interpolation clockwise	G03	circular interpolation counter-clockwise
G04	dwell (pause)	G17/18/19	XY / ZX / YZ plane selection
G20 / G21	inch / metric input	G28	return to reference point
G40	cutter compensation cancel	G41 / G42	cutter comp. left / right
G43 / G49	tool length comp. on / cancel	G54–G59	work coordinate systems
G80	cancel canned cycle	G81–G89	canned drilling cycles
G90 / G91	absolute / incremental	G94 / G95	feed per min / per rev

Important M codes (miscellaneous)

Code	Function	Code	Function
M00	program stop	M01	optional stop
M02	end of program	M03 / M04	spindle clockwise / counter-clockwise
M05	spindle stop	M06	tool change
M08 / M09	coolant on / off	M30	end of program & rewind

11. NC Nodes (Tool positioning, interpolation)

- **Nodes** = the points (end points of tool-path segments) that define the tool path; the controller joins them by interpolation.
- **Linear interpolation (G01)** – straight line between two nodes. **Circular interpolation (G02/G03)** – arc given by end point + centre (I, J) or radius R.
- Direction of arc: clockwise (G02) when looking from the +ve side of the axis perpendicular to the plane; **I, J** are the incremental distances from the start point to the arc centre in X and Y.

12. Canned Cycles

- **Canned cycle** = a single G code (with parameters) that performs a **frequently repeated multi-step operation** such as drilling, boring, tapping, so that program length is reduced.
- Typical drilling cycle sequence: rapid to R-plane → feed to depth Z → (dwell / peck) → rapid retract.
- Format: G8x X_ Y_ Z_ R_ Q_ P_ F_ ; (Z = depth, R = rapid-return (reference) plane, Q = peck depth, P = dwell time).

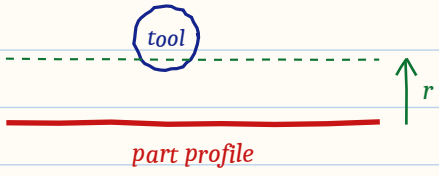
Code	Cycle	Use
G81	simple drilling	through holes, centre drill
G82	drilling with dwell	spot facing, counter-bore
G83	peck drilling (deep hole)	deep holes, chip breaking
G84	tapping	threaded holes
G85 / G86	boring (feed in – feed out / spindle stop)	finish bores
G73	high-speed peck	quick chip break

- Cancel the cycle with G80; the X, Y positions of the following blocks repeat the cycle at every position.
- On a **lathe**: G90 (turning cycle), G92 / G76 (threading cycle), G71 / G72 (rough turning / facing), G70 (finishing), G74 / G75 (grooving / peck drilling).

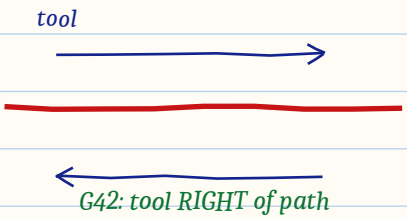
13. Cutter Radius Compensation (CRC / Cutter Diameter Compensation)

- The tool path programmed is that of the **part profile**; the control shifts the tool centre by the **tool radius** so that the edge cuts the programmed profile. Without CRC the programmer must calculate an offset path for every tool.

programmed path (tool centre) offset = r



G41: tool LEFT of programmed path



- **G41** = cutter compensation **left** (tool is to the left of the work profile in the direction

of travel – climb milling for a clockwise spindle).

- **G42** = cutter compensation **right**.
- **G40** = cancel compensation. The offset value is stored in the offset table (D01 etc.), so the same program works for a worn or re-ground cutter by just changing the offset.
- Steps: (1) approach with G00 away from the part, (2) turn on G41 / G42 with D-word in a **lead-in move** (a straight move at least equal to the radius), (3) machine the profile, (4) cancel with G40 in a lead-out move.
- Benefits: allows use of different cutter sizes, roughing and finishing with the same program, tolerance adjustment through offset change.

14. Tool Length Compensation

- Each tool has a different length; the control must know the **tool-length offset** from the spindle gauge line to the tool tip so the Z zero is correct for every tool.
- **G43** = tool length compensation **positive** (add H offset), **G44** = negative, **G49** = cancel. The H-number selects the stored value (H01 for T01).

```
G43  H01  Z100.0 ;      (tool 1 length is added to Z)
```

- Offsets are measured on the machine (touch-off or tool presetter) – the program need not change when the tool length changes. On lathes the tool offsets (X wear, Z wear) are called with the T word (T0101).

15. Manual Programs – Simple Components

(a) Milling – profile of a 60 × 40 rectangular block with a 10 mm hole at the centre (depth 10 mm)

- Work zero at lower-left corner top surface; tool: 10 mm end mill T01 for contour; 8 mm drill T02. Spindle 1000 rpm, feed 120 mm/min.

```
O1001 ;
N10 G21 G90 G17 G54 ; (metric, absolute, XY plane, work zero)
N20 T01 M06 ; (load end mill)
N30 S1000 M03 ; (spindle on CW)
N40 G00 X-10 Y-10 Z5 ; (rapid to start, above the work)
N50 G01 Z-10 F60 ; (plunge to depth)
N60 G41 D01 G01 X0 Y0 F120 ; (cutter comp left, lead-in)
```

N70 X60 ; N80 Y40 ; N90 X0 ; N100 Y0 ; (profile)

N110 G40 G00 X-10 Y-10 ; (cancel comp)

N120 Z50 M05 ; (retract, stop spindle)

N130 T02 M06 ; N140 S1200 M03 ;

N150 G81 X30 Y20 Z-12 R2 F80 ; (drill hole at centre)

N160 G80 G00 Z50 M05 ; N170 M30 ;

(b) Turning – step shaft: 40 mm bar, turn 30 mm dia for 40 mm length, then 20 mm dia for 20 mm (tool tip at X42 Z2)

O2001 ;

N10 G21 G90 G99 ; N20 T0101 ; N30 G96 S120 M03 ; (constant surface speed)

N40 G00 X42 Z2 ; N50 G01 X30 Z0 F0.15 ; N60 Z-40 ;

N70 X20 ; N80 Z-60 ; N90 X42 ; N100 G00 X100 Z100 ; N110 M30 ;

(c) Circular interpolation

- Quarter arc from (20, 0) to (0, 20) with centre at origin, counter-clockwise:

G03 X0 Y20 I-20 J0 F100 ; (or G03 X0 Y20 R20)

(d) Drilling a pattern of 4 holes using a canned cycle

G81 X20 Y20 Z-15 R2 F90 ; X60 ; Y60 ; X20 ; G80 ; (the cycle repeats at each position)

Computer Assisted Part Programming – APT

16. Computer-Assisted Part Programming

- In **computer-assisted programming** the programmer writes the program in a **high-level, English-like language**; the computer calculates cutter-location (CL) data and a post-processor makes the machine code. Used for complex 3D / 5-axis parts. Languages: **APT** (Automatically Programmed Tool, MIT 1956), ADAPT, EXAPT, COMPACT II, Pro-NC, and modern CAM (Mastercam, NX CAM).
- Four stages of APT processing: **(1) Input translation** (source program → machine-readable), **(2) Arithmetic (definition and motion) calculations**, **(3) Editing / CL file output (CLDATA)**, **(4) Post-processing** → NC program for a particular machine and control.
- The **part programmer** writes the part program; the **processor** (general, machine-independent) generates CLDATA; the **post-processor** (machine specific)

outputs G-code.

Source program → APT processor → CL file → Post-processor → NC part program → Machine.

17. APT Statements – the five types

(a) Geometry statements – define the elements of the part

- Format: symbol = geometry type / defining parameters.

P1 = POINT / 20, 30, 0 (point by coordinates)

L1 = LINE / P1, P2 (line through two points)

L2 = LINE / P1, PARLEL, L1 (parallel line) L3 = LINE / P1, PERPTO, L1

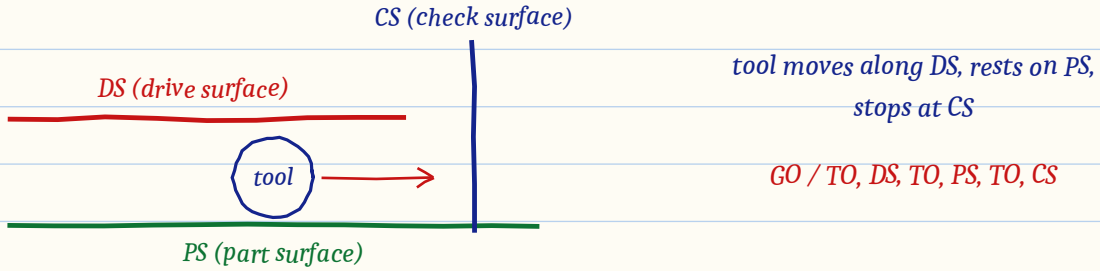
C1 = CIRCLE / CENTER, P1, RADIUS, 15 (circle)

PL1 = PLANE / P1, P2, P3 (plane through three points)

- Many forms are available: line by slope (LINE / P1, ATANGL, 45), circle tangent to lines (CIRCLE / YSMALL, L1, L2, RADIUS, 10), ellipse, cone, cylinder, sphere, vectors, matrices, patterns (PATERN).

(b) Motion statements – tool movement

- **Point-to-point**: FROM / P0 (start point); GOTO / P1 ; **GODLTA / dx, dy, dz** (incremental move).
- **Continuous path**: uses three surfaces – **drive surface (DS)**, **part surface (PS)** and **check surface (CS)**. The tool is kept tangent to the DS and PS and stops at the CS.
- Start-up: **GO / TO, DS, TO, PS, TO, CS** ; then moves: **GOLFT / L1, TANTO, C1** (go left along L1 tangent to C1), **GORGT, GOFWD, GOBACK, GOUP, GODOWN**.
- Tool direction terms: TO (tool touches surface), ON (tool centre on surface), PAST (tool beyond the surface), TANTO (tangent to).



(c) Post-processor statements – machine & control data

- **MACHIN / name, number** (select the post-processor: e.g. MACHIN / FANUC, 3).
- **COOLNT / ON, OFF, MIST, FLOOD**, **SPINDL / 1000, CLW, (CCLW)**, **FEDRAT / 120**, **RAPID**, **LOADTL / 1** (tool change), **TURRET**, **END**, **FINI** (end of the program), **CUTTER / 10** (tool diameter), **TOLER**, **INTOL / OUTTOL** (tolerances), **OFSTNO**.

(d) Auxiliary statements – supply non-geometric information

- **PARTNO** (title / name), **REMARK**, **INSERT**, **PRINT**, **CLPRNT**, **PPRINT**, **INTOL**, **OUTTOL**, **COOLNT**, **TLON**, **STOP**.
- Their position: PARTNO is the first statement; FINI is the last.

(e) Macro statements – subprograms (repeated shapes)

- **Macro** = a named group of APT statements with parameters that is **defined once and called many times** – like a subroutine.

HOLE = MACRO / X, Y, DEPTH (macro header with arguments)

GOTO / X, Y, 5 ; GODLTA / 0, 0, -DEPTH ; RAPID ; GODLTA / 0, 0, DEPTH ;

TERMAC (end of the macro)

CALL / HOLE, X = 20, Y = 30, DEPTH = 12 (call with actual values)

18. APT Program for a Simple Component

Example – profile milling of a plate (rectangle 80 × 50) and a drilled hole, cutter dia 10 mm

PARTNO PLATE (title)

MACHIN / FANUC, 3 (post-processor)

CLPRNT ; UNITS / MM ; CUTTER / 10 (print CL data, metric, tool dia)

P0 = POINT / -20, -20, 5 (tool start point)

L1 = LINE / 0, 0, 0, 80, 0, 0 (bottom edge) L2 = LINE / 80, 0, 0, 80, 50, 0

L3 = LINE / 80, 50, 0, 0, 50, 0 (top edge) L4 = LINE / 0, 50, 0, 0, 0, 0

PS = PLANE / 0, 0, 1, 0 (part surface z = 0) P5 = POINT / 40, 25, 0

FROM / P0 ; FEDRAT / 120 ; SPINDL / 1000, CLW ; COOLNT / ON

GO / TO, L1, TO, PS, TO, L4 (start-up statement)

GORGT / L1, PAST, L2 ; GOLFT / L2, PAST, L3 ; GOLFT / L3, PAST, L4 ; GOLFT / L4, PAST, L1

GOTO / P0 ; RAPID ; GOTO / P5 ; GODLTA / 0, 0, -12 ; GODLTA / 0, 0, 12 ; SPINDL / OFF ;

COOLNT / OFF

FINI

- **Result**: processor creates CL data → post-processor converts into G / M codes for the

selected machine.

Advantages of APT / computer-assisted programming

- Easy for complex geometry (3D, 5-axis), less error, calculations done by computer, tool path plot / simulation, reusable macros, one program for many machines (different post-processors).

Limitations

- Needs a computer and a trained programmer, geometry has to be defined in the language, cost of software, no direct link to a CAD model (older systems) – solved in modern CAD/CAM.

19. Quick Revision

NC elements: part program + MCU + machine tool. NC types: PTP, straight-cut, contouring. Loops: open (stepper) / closed (servo + feedback).

- CNC = microcomputer inside MCU; DNC = host computer controls many machines; adaptive control = ACC, ACO, GAC.
- NC word = letter + number; block = line; modal vs non-modal; G = preparatory, M = miscellaneous.
- Canned cycles G81–G89 reduce program length; G41/G42/G40 cutter radius compensation; G43/G49 tool length compensation.
- APT: 5 statement types – geometry, motion, post-processor, auxiliary, macro. Motion uses DS, PS, CS.

Likely exam questions

- Define NC and explain the elements of an NC system with a block diagram.
- Differentiate between open-loop and closed-loop NC systems. Explain NC, CNC and DNC.
- Describe the features and types of machining centres.
- What is adaptive control? Explain its types with applications.
- Explain NC word, block format and the commonly used G and M codes.
- Write a manual part program (with canned cycle / compensation) for the given component.
- Explain geometry, motion, post-processor, auxiliary and macro statements of APT with examples.

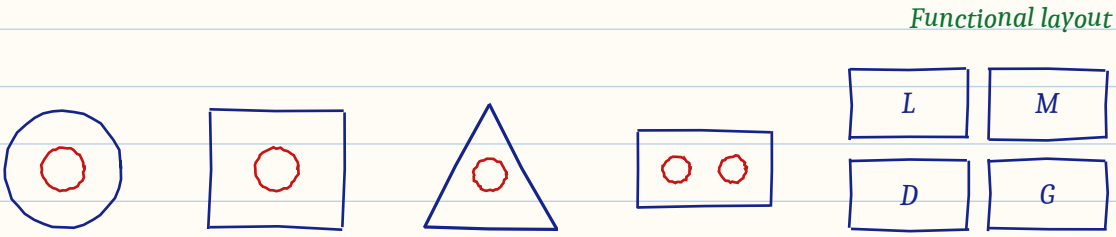
UNIT – IV : Group Technology, FMS & CAQC

1. Group Technology (GT) – Introduction

- **Group technology** is a manufacturing philosophy in which **similar parts are identified and grouped together** (into part families) to take advantage of their **similarities in design and manufacturing**.
- Idea: instead of treating each part as new, solve a problem once for the whole family (similar design, tooling, process plan, setup). Used in batch production (50 to 75% of all manufacturing is batch type).
- Similarities: **design attributes** (shape, size, material, tolerance) and **manufacturing attributes** (sequence of operations, machines, tools, fixtures).

2. Part Family

- **Part family** = a collection of parts that are similar either in geometric shape and size or in the processing steps required to make them.
- Parts with different shapes may belong to the same family if they need the same operations (e.g. turning → drilling → milling → grinding).



different shapes, similar processes: turn → drill → mill → grind = ONE PART FAMILY

- Advantages of identifying families: standard process plans, design retrieval (avoid duplicate designs), reduced setups, cell formation.

Methods of forming part families

- **Visual inspection** (ocular) – parts or photographs sorted by looking at them; simple but not very accurate.
- **Production flow analysis (PFA)** – uses the **route sheets**: machines needed by each part are analysed and parts with the same machine sequence are grouped.

- **Classification and coding systems** – every part is given a code number based on its design and manufacturing features; parts with similar codes form a family. (most widely used)

3. Classification and Coding

- **Classification** = grouping of parts into classes by similarity. **Coding** = assigning symbols (numbers / letters) that represent the part's features.

Types of code structure

- **Hierarchical (monocode)**: each digit depends on the meaning of the previous digit (tree) – compact, detailed for one aspect; difficult to design.
- **Chain (polycode / attribute)**: each digit has its own fixed meaning independent of other digits – easy to use; longer code.
- **Mixed (hybrid) type**: combination of both – most commercial systems (Opitz, MICLASS).

Features of a good coding system

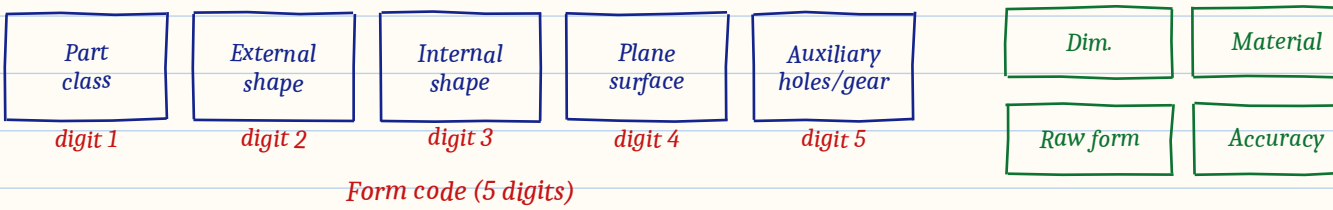
- Based on both design and manufacturing data, easy to learn and use, adequate length, flexible, covers all parts of the company (universal), supports computer storage.

Popular systems

System	Origin	Remark
Opitz	Prof. Opitz, Aachen (Germany)	9-digit; best known; mixed code
MICLASS	TNO, Netherlands	12 digit + 12 optional; computer interactive
CODE	Manufacturing Data Systems Inc, USA	8 digit; hexadecimal; for company use
DCLASS	Brigham Young University	tree-structured; user can design
KK-3	Japan	21 digits for large part range

The Opitz system (9 digits)

- **Form code (5 digits)**: digit 1 – part class (rotational L/D ratio, non-rotational); digit 2 – external shape & elements; digit 3 – internal shape; digit 4 – plane surface machining; digit 5 – auxiliary holes and gear teeth.
- **Supplementary code (4 digits)**: dimension (size), material, original shape of raw material, accuracy.
- A fourth group (special code, optional) for company use.



Example: Opitz code 1 0 1 0 0 | 2 1 3 2 (short rotational part)

- *Digit 1 (part class): 0 → rotational, L/D ≤ 0.5 (disc); 1 → 0.5 < L/D < 3; 2 → L/D ≥ 3 (long shaft); 3, 4 → rotational with deviations (e.g. offset holes, ribs); 5 → special rotational; 6, 7, 8 → non-rotational (flat, long, cubic); 9 → special non-rotational.*

4. Production Flow Analysis (PFA) and Cell Formation

- *Starts with the **machine–part incidence matrix**: rows = machines, columns = parts; entry 1 if the part needs that machine.*
- ***Rank Order Clustering (ROC)**: read each row (and column) as a binary number; sort rows in decreasing order of value, then sort columns, repeat until no change → blocks of 1's along the diagonal = **machine groups (cells) and part families**.*

Example – 4 machines × 5 parts

Machine	P1	P2	P3	P4	P5
M1	1	0	0	1	0
M2	0	1	1	0	1
M3	1	0	0	1	0
M4	0	1	1	0	0

- *Row values (weights 2^4 … 2^0 for P1…P5): M1 = 10010 = 18; M2 = 01101 = 13; M3 = 18; M4 = 01100 = 12 → order M1, M3, M2, M4.*
- *After re-ordering columns by their values: **cell 1 = {M1, M3} making {P1, P4}**; **cell 2 = {M2, M4} making {P2, P3, P5}** – a clean block diagonal matrix, no inter-cell movement.*

If a few "1"s fall outside the blocks (exceptional parts / bottleneck machines), duplicate the machine or subcontract the part.

5. Advantages and Limitations of Group Technology

Advantages (benefits)

- **Design**: variety reduction, standardisation, design retrieval, less new drawings, easier cost estimation.
- **Process planning**: standard process plans for a family, faster planning, **less setup time** through similar setups.
- **Production**: reduced material handling, work-in-process, lead time; better machine utilisation; simple scheduling.
- **Quality & cost**: fewer defects, lower scrap, reduced tooling and inventory, higher job satisfaction (team work in cells).

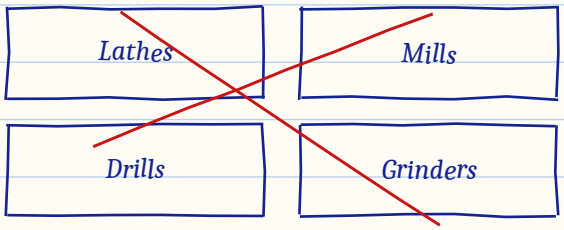
Limitations

- Cost and time of **classifying and coding** all old parts; cost of re-arranging the layout; management and worker attitude (cross-training); loss of flexibility if product mix changes; machines may be under-used within a cell; not suited to very large or very small variety.

6. Group Technology Machine Cells

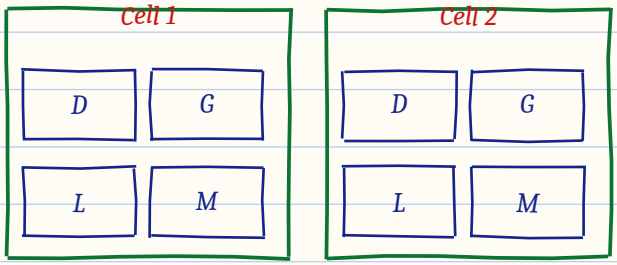
- **GT cell (cellular manufacturing)** = group of **dissimilar machines** put together (in a cell) to produce one part family; parts move in small batches within the cell.

(a) Process / functional layout



long, crossing material flow

(b) GT cells (machine groups)



short, smooth flow

Types of GT cell layouts

- **Single machine cell**: one machine with fixtures / tools for a family (e.g. a machining centre).
- **Group machine cell, no material handling** – machines grouped but parts moved by operators manually (job-shop like).
- **Group machine cell with in-line (flow-line) material handling**: machines arranged in the processing sequence with a conveyor / robot; or **U-shaped layout** (operator

inside the U).

- **Cluster / loop layout**: loop conveyor around machines for varying sequences; **flexible manufacturing cell** (FMC) – CNC machine + pallet + robot.
- Layout types by flow: **in-line, U-shaped, loop, ladder, open-field**.

Benefits of GT cells

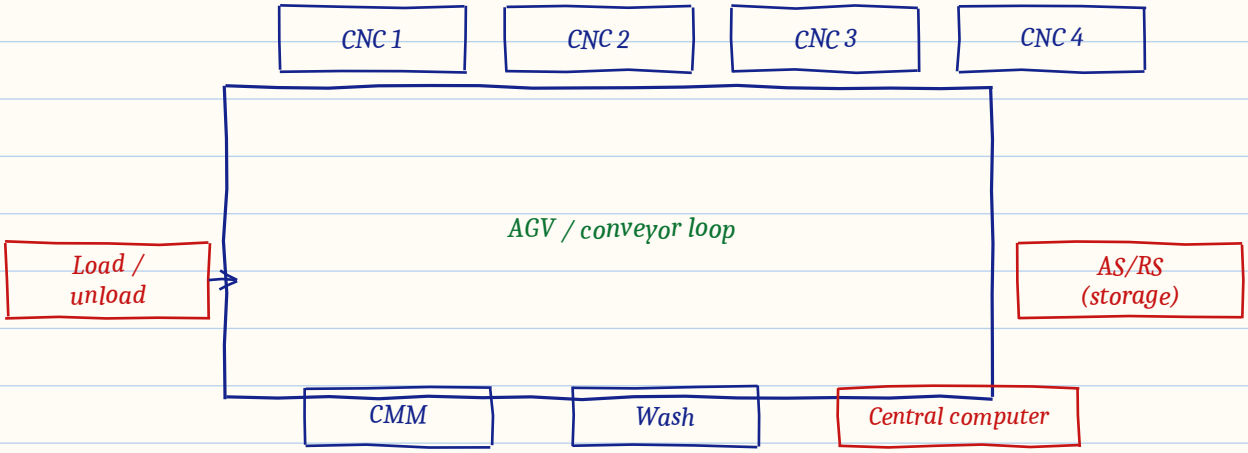
- Shorter material flow and lead time, lower WIP, easy supervision, better quality ("owned" by the team), less setup time, better use of floor space, easy automation → basis of **FMS**.

7. Flexible Manufacturing System (FMS) – Introduction

- **FMS** = highly automated GT machine cell with a group of **processing workstations (CNC machines)** interconnected by an **automated material handling & storage system** and controlled by a **distributed computer system**.
- "Flexible" because it can process **different part styles simultaneously** and the production schedule can be changed (in mid-batch) – no or little changeover time between parts.
- Flexibility tests: (1) ability to process different part styles without batch changeover, (2) quick changeover of the production schedule, (3) ability to cope with equipment breakdown, (4) ability to accommodate new part designs.
- Types of flexibility: **machine, process, product, routing, volume, expansion, production** flexibility.
- Positioning: between high-variety low-volume job shop and high-volume transfer lines – medium volume, medium variety (4 to 100 part types, 20 to 20,000 pieces/year).
- **Types of FMS**: by number of machines – single machine cell (FMC), flexible manufacturing system (3+ machines), flexible transfer line; by layout – in-line, loop, ladder, open field, robot-centred cell; by flexibility – **dedicated FMS** (fixed range of parts, sequential) vs **random-order FMS** (wide range of parts, any routing).

8. Components of an FMS

- **1. Processing (work) stations** – CNC machining centres, turning centres, also inspection (CMM), washing, assembly, sheet-metal press stations. Machines have ATC, tool magazines and probes.
- **2. Material handling and storage system** – moves parts, pallets, tools between stations and stores them.
- **3. Computer control system** – central computer + microprocessors at the stations; controls and coordinates everything.
- **4. Human labour** – load / unload, tool change and setting, programming, maintenance, system management.



9. Material Handling Systems in FMS

- Functions: random, independent movement of workparts between stations; handle a variety of part configurations; temporary storage; convenient load / unload; compatibility with computer control; access to machines for maintenance.

Primary handling systems

- **Automated Guided Vehicles (AGV)**: battery-driven carts following a wire / optical / laser guide path; flexible routes, good for loop & open-field layouts; carry pallets with fixtures.
- **Conveyors** (roller, belt, chain, overhead): good for in-line layout; fixed paths; low cost.
- **Rail-guided vehicles (RGV)**: vehicles on rails – fast, accurate, for in-line layouts.
- **Industrial robots** – load/unload machines; gantry robots for in-line / cell layouts.

Secondary / supporting handling

- **Pallets and fixtures** – the workpiece is clamped on a pallet fixture, standard interface to the machine (pallet changer).
- **AS/RS (Automated Storage and Retrieval System)** – computer-controlled racks with stacker cranes for raw material, WIP, tools.
- **Tool handling**: tool magazines, tool carriers, tool presetting, tool-life monitoring; **chip & coolant handling**.

10. Computer Control System of FMS

- Hierarchical structure (host computer + station controllers + machine controls), usually with **DNC** capability.
- Functions:
  - **Workstation control** – CNC programs downloaded and run at each machine.
  - **Distribution of control instructions** (part programs) to stations – DNC.
  - **Production control** – part-mix, rate of launching, scheduling, dispatching.
  - **Traffic control** – routing of AGVs / conveyors, collision avoidance.
  - **Shuttle (pallet) control**, work-handling and load / unload scheduling.
  - **Work-piece monitoring** (status of every part), **tool control** (location, tool life, replacement), **performance monitoring and reporting**, **diagnostics** and failure recovery.
- Software: real-time operating system, simulation to test the schedule, database of parts / tools / programs.

11. Advantages (and Limitations) of FMS

Advantages

- Higher machine utilisation (up to 80 to 90%) – parts continue while machines idle in job shop.
- Reduced work-in-process, inventory and floor space (fewer machines).
- **Shorter manufacturing lead time**; faster response to market change; batch size of one is possible.
- Lower direct labour (lights-out operation); consistent quality, lower scrap and rework.
- Flexibility in scheduling and part mix; ability to continue production in case of

breakdown (re-routing).

Limitations

- Very high capital investment; complex planning and software; need skilled people; limited to a **family** of parts; long development time; downtime of the central computer can halt everything.

12. Computer Aided Quality Control (CAQC) – Terminology

- **Quality** = fitness for use / conformance to specifications. **Quality control (QC)** = operations used to see that the product meets the quality standard; **quality assurance (QA)** = planned system of activities to give confidence.
- **Inspection** = checking (measuring, gauging) a product against the specification to decide **accept or reject**. **Testing** = determining performance / strength / life by applying loads or conditions (destructive or non-destructive).
- **Tolerance**: permitted variation of a dimension. **Accuracy**: closeness to true value. **Precision**: repeatability. **Resolution** (least count), **sensitivity**, **calibration** (comparison with a standard).
- **SPC** (statistical process control) – control charts ($\bar{X}$ – R chart), process capability Cp, Cpk; **sampling inspection** (AQL) vs **100% inspection**.
- **TQM** (total quality management), **Six Sigma** (3.4 defects per million), **zero defects**, **cost of quality** (prevention, appraisal, internal and external failure).
- **Quality control activities**: incoming materials inspection, in-process inspection, final inspection, gauge calibration, defect analysis, corrective action.

13. Inspection and Testing

- Inspection can be **off-line** (in a quality lab, after manufacture), **on-line** (in-process, at the machine), **inline** (part of the production line) or **post-process**.
- **Manual inspection**: vernier calliper, micrometer, dial gauge, go / no-go gauges, height gauge, surface plate – slow, operator-dependent.
- **Automated inspection**: sensors and computer to measure, decide and feed back to the process – faster, 100% inspection, consistent, data recorded for SPC.
- Methods: **contact** methods (a probe touches the part) and **non-contact** methods (optical and non-optical sensing) – used under CAQC.
- **Testing**: destructive (tensile, hardness, impact, fatigue) and **non-destructive testing**

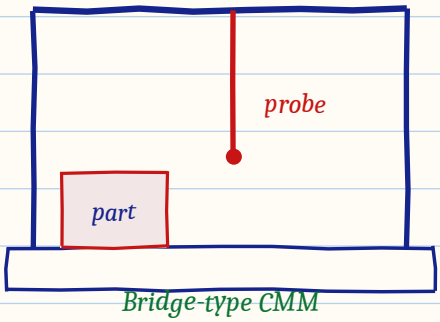
(NDT): dye penetrant, magnetic particle, eddy current, ultrasonic, radiography.

14. Contact Inspection Methods

- A physical **probe touches the surface** and the position is read → used in **CMM** and in gauges on the machine tool.

Coordinate Measuring Machine (CMM)

- A machine with a **touch probe** moved along X, Y, Z axes; the computer reads the coordinates of the touched points and computes dimensions, form, position, orientation.



Bridge-type CMM

X – bridge
Y – table
Z – quill (probe)

Probe touches the part → computer records X, Y, Z and calculates size, position, form

- **Types of CMM structure**: **bridge**, **cantilever**, **gantry**, **horizontal-arm**, **column / articulating-arm (portable)**.
- **Probes**: **touch-trigger (hard) probe** gives a signal when contact is made; **analog (scanning) probe** measures continuously; styli of ruby ball; non-contact laser probes on some CMMs.
- Modes: manual, **CNC (DCC – direct computer control)** program-driven measurement; programs can be generated from CAD models (off-line).
- Software: measures lines, circles, planes, cylinders, flatness, roundness, GD&T; statistical reports; **reverse engineering** (digitising a model).
- Advantages: high accuracy (µm), flexible, no special gauges, fast for complex parts, result automatically available to the computer. Limits: slow compared with non-contact, probe pressure on soft parts, environment (temperature) control needed.

Other contact methods

- **Touch trigger probes on CNC machines** – measure workpiece, set work offset, tool length & breakage (in-process).
- **Linear variable differential transformer (LVDT)**, dial indicators with electrical

output, **air gauges** (pneumatic, no or light contact), **electronic gauges** for automatic sorting machines.

15. Non-contact Inspection Methods

(a) Optical methods

- **Machine vision**: a camera + lighting + computer image processing; image is digitised into pixels and analysed for dimensions, presence / absence, defects, orientation, reading codes. Steps: image acquisition → pre-processing → segmentation → feature extraction → decision.
- **Laser scanning / laser triangulation**: laser beam sweeps the part; position of reflected spot measures distance – gives 3D surface data (point cloud) quickly.
- **Profile projector / optical comparator** (shadow image magnified on a screen), **toolmaker's microscope**, **optical interferometry** for surface flatness, **laser micrometer** (shadow of a part cuts a laser beam), **structured light / photogrammetry**.
- Advantages: fast, no contact force or wear, 100% inspection, small and soft parts. Limits: affected by lighting, surface reflectivity and dirt.

(b) Non-optical methods

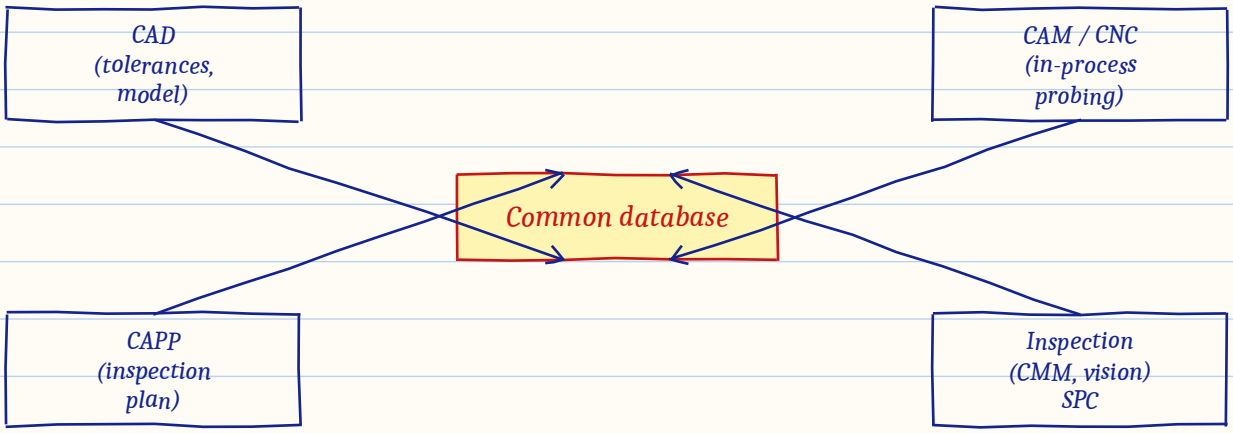
- **Electrical**: capacitive and inductive (eddy current) proximity sensors measure gap, thickness, coating.
- **Pneumatic (air) gauging**: back-pressure of air jet is a measure of the gap – high accuracy (0.5 µm).
- **Ultrasonic**: sound pulse echoes measure thickness and internal defects. **Radiography (X-ray / gamma)**: internal defects, castings and welds. **Magnetic** and **eddy-current** testing for cracks. **Acoustic emission**, **thermography**.
- **Tactile / capacitive** sensing in automatic sorting & **in-process gauging** on grinders.

16. Integration of CAQC with CAD and CIM

- **CAD ↔ CAQC**: the CAD model gives dimensions and **GD&T tolerances**; an inspection program (e.g. for the CMM) is generated automatically from the model; measured data are compared with the CAD model (deviation colour map) → **inspection planning by CAD**.
- **CAQC ↔ CAM / CNC**: probing and in-process gauging on the CNC machine →

feedback to correct tool offsets, tool wear compensation; closed-loop quality control.

- **CAQC ↔ CAPP**: process plans include inspection steps, frequencies, gauges.
- **CIM**: all quality data (inspection results, SPC charts, defects, calibration) are stored in the **common database**; used by planning, scheduling and management; **information flow in both directions**.



Benefits of integrated CAQC

- Quality built in (prevention, not detection), shorter inspection time, 100% inspection possible, **real-time process correction**, traceability, lower scrap and cost of quality, supports Six Sigma and ISO 9000.

17. Quick Revision

GT = group similar parts → part family → machine cell. FMS = GT cell + CNC machines + automated handling + computer control.

- Family forming: visual, PFA, classification & coding (Opitz 9 digit: 5 form + 4 supplementary).
- Code types: hierarchical, chain, mixed. ROC algorithm for cell formation. GT benefits: less setup, WIP, lead time.
- FMS components: workstations, material handling (AGV, conveyors, robots, AS/RS), computer control, people.
- CAQC: contact (CMM, probes) and non-contact (optical – vision, laser; non-optical – ultrasonic, eddy current, pneumatic).
- Integration: CAD gives tolerance data, CAM gets feedback, all data in CIM database.

Likely exam questions

- What is group technology? Explain part family, classification and coding with the Opitz system.
- Describe the advantages and limitations of GT. Explain the types of GT cell layouts.
- Define FMS. Explain the components of an FMS with a neat sketch.
- Explain the material handling systems and the computer control system of an FMS.
- Explain CMM and its types. Discuss contact and non-contact inspection methods.
- How is CAQC integrated with CAD and CIM?

UNIT – V : CAPP, Production Planning & Trends

1. Computer Aided Process Planning (CAPP)

- **Process planning** = deciding **how** a part will be made: selection of **processes, machines, tools, fixtures, cutting data, operation sequence and times** – result is the **route sheet / operation sheet** (process plan). It links **design and manufacturing**.
- Manual process planning depends on the **experience of the planner** → inconsistent, slow, skilled labour needed.
- **CAPP** = use of the computer to prepare the process plan automatically (or semi-automatically).

Steps in process planning

- Study the drawing → select raw material → select processes and sequence → select machines → select tools, fixtures, gauges → fix cutting conditions (speed, feed, depth) → estimate time & cost → prepare the route sheet / NC program.

Benefits of CAPP

- Standardised and **consistent** plans, **faster planning**, less skill needed; reduces planning staff and lead time.
- Better use of the shop's resources, lower manufacturing cost, links with **CAD / CAM / MRP / NC programming**, easy updating of plans, supports GT and FMS, possibility of optimum plans.

2. Retrieval Type CAPP (Variant approach)

- Based on **Group Technology**: parts are classified and coded into **part families**; a **standard process plan (master plan)** is stored for each family.
- Procedure: code the new part → find its family → **retrieve the standard plan** → **edit / modify** it for the particular part → print the route sheet.

RETRIEVAL (VARIANT) CAPP



standard plans stored in database → retrieved & modified → new plan

- Two stages: **preparatory stage** (building the database: coding, grouping, creating standard plans) and **production stage** (using the system).
- Popular systems: **CAPP (CAM-I)**, MIPLAN, MultiCAPP, CPPP.

Advantages

- Simple, quick to develop, low cost, easy to learn; the planner keeps control; good where parts are similar and a large number of families exist.

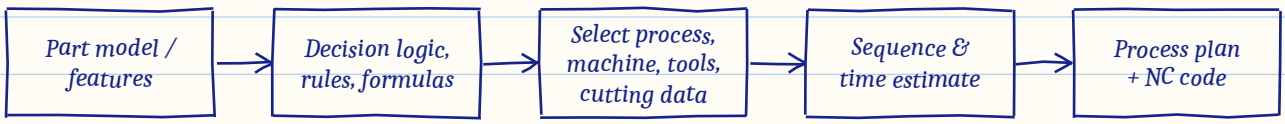
Disadvantages

- Needs a **GT coding** effort; plans are only for existing families; new / dissimilar parts need manual planning; quality depends on standard plans and the planner who edits; cannot optimise.

3. Generative Type CAPP

- Process plan is **generated automatically** from scratch for each new part using **decision logic (rules), formulas, technology data and algorithms** – no standard plan is retrieved.
- Input: complete description of the part – a **CAD model with features** (holes, slots, steps, tolerances, surface finish, material) – plus manufacturing knowledge in the computer.

GENERATIVE CAPP



plan is created fresh by the computer using logic – no standard plan needed

- **Knowledge base**: machine capabilities, tooling database, machinability data, cost / time formulas and the **decision tables / decision trees / expert-system rules** (IF feature = hole of Ø10, depth 30 THEN drill with a 10 mm drill).

- Techniques: decision trees & tables, **expert systems (AI)**, feature recognition from CAD, **process-capability matching**.
- Popular systems: AUTAP, **CIMS/PRO**, XPLAN, **GARI**, Part-Tec.

Advantages

- Fully automatic, **consistent and optimised** plans; handles new parts; direct link to CAD and NC programming; fast for complex parts; no GT coding needed.

Disadvantages

- Very difficult and costly to develop (process knowledge must be converted to logic); needs **feature-based CAD**; maintenance of rules; **semi-generative** (hybrid) systems combine both approaches in practice.

Retrieval vs generative

Feature	Retrieval (variant)	Generative
Basis	standard plan of a part family	logic / rules and the CAD model
GT code needed	yes	not necessary
New part	needs modification by the planner	plan generated automatically
Development	easy, cheap	difficult, expensive
Optimisation	limited	possible
Use	repetitive, similar parts	complex, varied parts; CAD / CAM integration

4. Machinability Data Systems

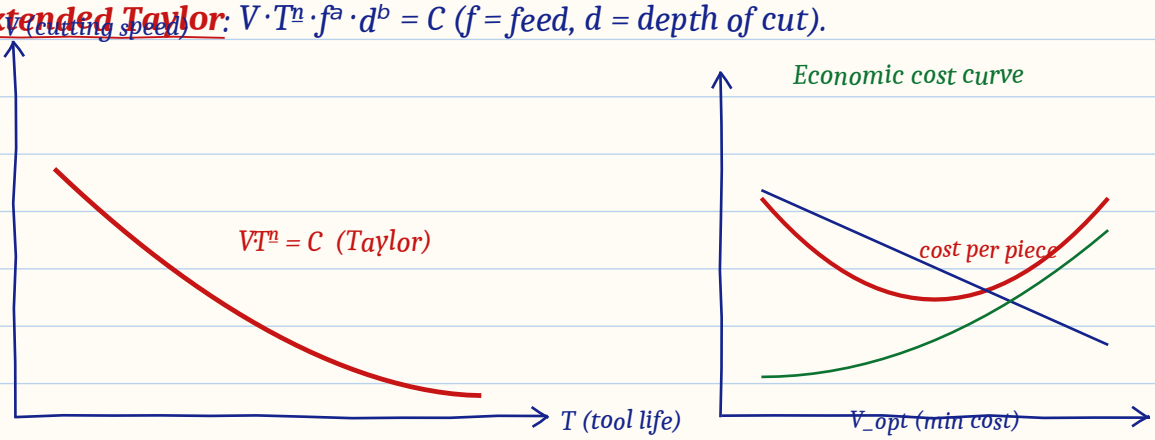
- **Machinability** = ease with which a material can be machined (tool life, surface finish, cutting force, power, chip form). Machinability data = recommended **cutting speed, feed, depth of cut, tool material and geometry, coolant** for a given work material and operation.
- **Machinability data system** = computerised database + models that supply the optimum cutting conditions to the process planner / NC programmer.
- Two forms: **database (look-up tables)** from handbooks and experiments, and **mathematical models** (Taylor's equation) calculating the optimum values.

$$V \cdot T^n = C \quad \text{(Taylor's tool life equation)}$$

- V = cutting speed (m/min), T = tool life (min), n = exponent (HSS 0.1 to 0.15, carbide

0.2 to 0.5, ceramic 0.4 to 0.7), C = constant.

- **Extended Taylor**: $V \cdot T^n \cdot f^a \cdot d^b = C$ (f = feed, d = depth of cut).



- **Optimum cutting speed** may be selected for (a) **minimum cost** per piece, (b) **maximum production rate** (minimum time), or (c) **maximum profit**. Min-cost speed is lower than max-production-rate speed.

$$T(\text{min cost}) = (1/n - 1)(Tc + C_t/C_m)$$

- Tc = tool change time, C_t = tool cost per cutting edge, C_m = machine + labour cost per minute.

$$T(\text{max production}) = (1/n - 1) \cdot Tc$$

- **Example**: for carbide n = 0.25, Tc = 2 min: T(max production) = (4 – 1) × 2 = 6 min.
- Data are obtained from experiments, handbooks (Machining Data Handbook), manufacturers; examples of systems: **MACHDATA**, **METCUT**, **Machinability Data Center (MDC)**, **Kennametal Novo**.
- Used in CAPP, NC programming and adaptive control; **advantages**: optimal cutting conditions, less trial and error, longer tool life, lower cost, shorter cycle time.

5. Computer Generated Time Standards

- **Time standard** = the time allowed to a qualified, trained worker working at normal pace to complete one operation (or job) – used for **scheduling, costing, wage incentives, capacity planning, plant layout**.
- Conventional method: stop-watch time study and work sampling – needs the job to be present and an observer.
- **Computer-generated time standards** (standard data systems / predetermined time systems): a computer calculates the **standard time from the process plan** before the

part is made – no stop-watch.

Methods

- **Standard data method** – stored time values for elements (e.g. rapid traverse, load / unload, tool change) and **formulae for machining time** with cutting data.
- **Predetermined motion time systems (PMTS)** – MTM (Methods-Time Measurement), **MOST**, work factor: basic human motions are given fixed time values and summed.
- Programs: **MTM-based software**, **COMTEK**, **MTM-LINK**, **AUTOPLAN / CAPP time module**.

Computation of the standard time

- **Time per piece Tc = Tsu / Q + Tm + Th + Tt** → setup time per batch (Tsu / Q), machining time Tm, work handling time Th, tool handling time Tt.

$$Tm\ (turning) = L / (f \cdot N)\ \ min, \quad where\ N = 1000\ V / (\pi\ D)\ \ rpm$$

- **Standard time = Normal time × (1 + allowances)** – allowances for personal needs, fatigue and delay (usually 10 to 20%).
- **Example**: Turning a Ø50 mm bar, length 120 mm, V = 100 m/min, f = 0.2 mm/rev: N = 1000 × 100 / (π × 50) = 637 rpm; Tm = 120 / (0.2 × 637) = **0.94 min**; with handling 0.6 min + allowance 15% → standard time = (0.94 + 0.6) × 1.15 = **1.77 min / piece**.

Advantages

- Fast and consistent, available at the planning stage, no disturbance to the workers, easy updating when methods change, integrates with CAPP, MRP and costing.

Computer Integrated Production Planning

6. Production Planning and Control (PPC) – Overview

- **PPC** = planning, scheduling and controlling of production so that the right product is made in the **right quantity, at the right time, at the lowest cost**.
- Functions: forecasting, **aggregate planning**, **master production schedule (MPS)**, material requirements planning, **capacity planning**, scheduling & dispatching, shop-floor control, inventory control, purchasing, and cost control.
- **Computer integrated production planning (CIPP)** = these functions performed by computer systems sharing a common database; information from CAD (BOM), CAPP (routings, times) and the shop floor (status) feed the planning.

7. Capacity Planning

- **Capacity** = maximum output a facility (machine, work centre, plant) can produce per period (hours / week). **Capacity planning** = determine the labour and machine capacity needed to meet the production plan and balance it against the available capacity.
- Levels: **Resource requirements planning (long range)**, **Rough-cut capacity planning (RCCP – from MPS)** and **Capacity requirements planning (CRP – from MRP, medium / short range)**.

Steps

- Find the **load** (required hours) on each work centre from the planned orders and routings (set-up + run time) → compare with **available capacity** → find overload / underload → adjust.

Required hours = Σ (Tsu + Q × Tp) of all orders on the work centre

Utilisation = (hours used) / (hours available); Available capacity = machines × hours × efficiency × utilisation

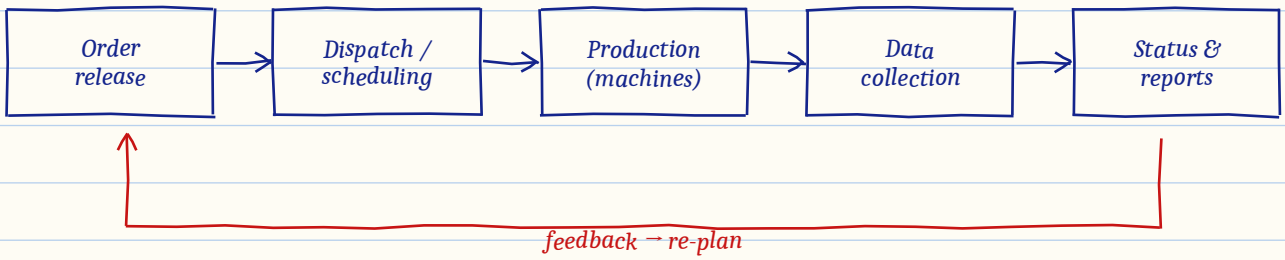
- **Example**: a work centre has 2 machines × 40 h/week × 85% efficiency = 68 h available. Orders need 80 h → overload 12 h.
- Ways to correct: overtime, extra shift, subcontracting, alternate machines / routing, rescheduling (level loading), add machines, hire labour. If capacity is too

high → **underload** → reduce shifts / take more orders.

- **Finite loading** (do not exceed capacity) vs **infinite loading** (load without limits, then adjust). Capacity strategies: lead, lag, match.
- **CAPP + MRP** supply the data (routing and standard times) for CRP – shows the use of computer integration.

8. Shop Floor Control (SFC)

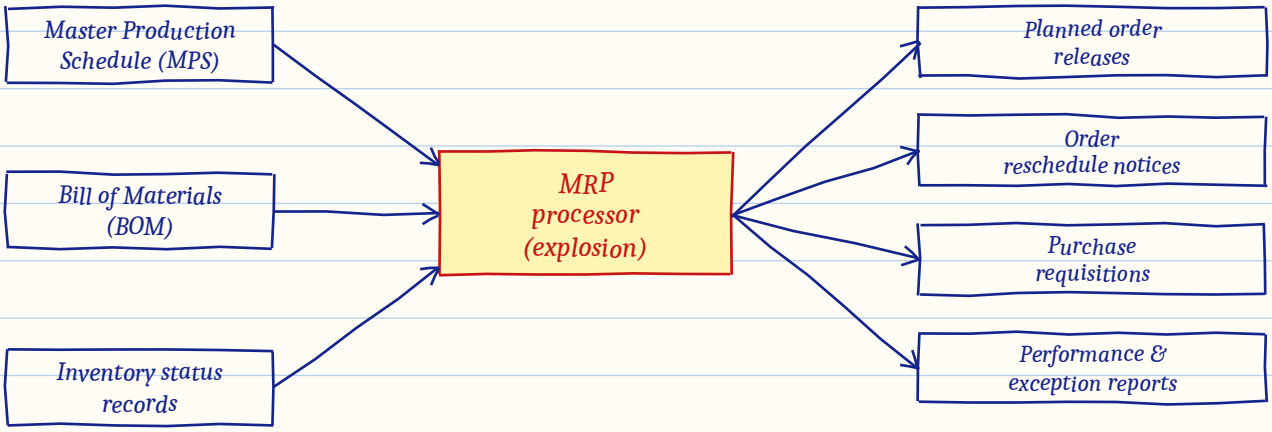
- **SFC** = system for **releasing orders to the shop, scheduling and dispatching them, and monitoring progress** on the shop floor; it follows after MRP. Provides the **feedback** to planning.
- Three phases: **(1) Order release** – authorise the production order with the documents (route sheet, drawings, material, tooling); **(2) Order scheduling and dispatching** – sequence the jobs at each work centre (priority rules like **FCFS, SPT, EDD, critical ratio**); **(3) Order progress** – monitor status, work-in-process, queue, exceptions.



- **Data collection (shop-floor data collection, SFDC)**: bar-code, RFID, **touch screens, machine monitoring, counters** – operators report start / stop, quantity, scrap; automated collection gives real-time status.
- Reports: work order status, **work-in-process**, **exception reports (late jobs)**, machine utilisation, scrap, **labour efficiency**.
- **Input / output control** and **priority control**; **MES (Manufacturing Execution System)** and **Kanban** are the modern forms.
- **Factory data collection systems** are the link of CIM between the shop floor and the planning system.

9. MRP – Material Requirements Planning (MRP-I)

- **MRP** = computer-based system that determines **what materials (components, sub-assemblies, raw materials) are needed, how many and when**, based on the master production schedule (MPS), the bill of materials and the inventory data.
- Applies to **dependent-demand items** (parts needed for end products), unlike reorder-point systems used for independent demand.
- **Inputs**: (1) **Master production schedule (MPS)** – what end items and when; (2) **Bill of materials (BOM)** – product structure (levels, quantity per parent); (3) **Inventory status file** – on-hand, on-order, lead times, lot-size rules, safety stock.



- **Outputs**: planned order releases, order reschedule notices (expedite / de-expedite), purchase requisitions, exception and performance reports.

MRP logic – the calculation (explosion)

- **Gross requirements** (from MPS or parent) – **scheduled receipts** – **projected on-hand** = **net requirements**.
- Net requirements are converted to **planned order receipts** using a **lot-sizing rule** (lot-for-lot, EOQ, fixed order quantity, period order quantity).
- **Planned order release** = planned receipt **offset by lead time**.

Net requirement = Gross requirement – (On hand + Scheduled receipts) + Safety stock

- Done level by level (low-level coding) from the end item down to the raw material – **BOM explosion**.

Example – MRP table (lot-for-lot, lead time 2 weeks, on hand 20)

Week	1	2	3	4	5	6
Gross requirement	0	0	40	0	60	30
On hand (end)	20	20	0	0	0	0
Net requirement	0	0	20	0	60	30
Planned order receipt	0	0	20	0	60	30
Planned order release	20	0	60	30	0	0

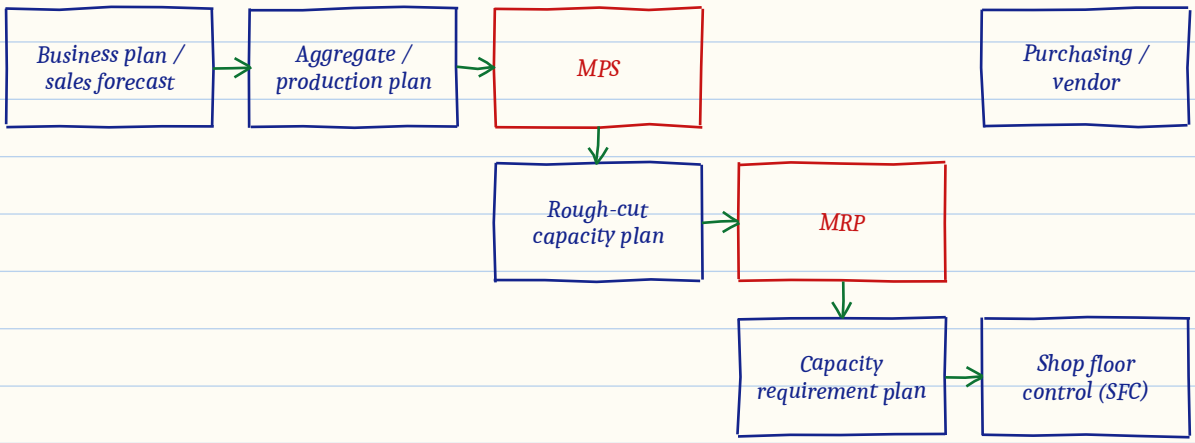
- Week 3: 40 – 20 on hand = 20 net → receipt in week 3, so the order is **released 2 weeks earlier (week 1)**; the 60 needed in week 5 is released in week 3, and the 30 in week 6 in week 4.

Benefits and limitations of MRP-I

- Benefits: lower inventory, fewer shortages, better delivery performance, priorities for the shop, planning of purchases.
- Limitations: **assumes infinite capacity** and fixed lead times; no feedback on capacity; accuracy of data (BOM, inventory) is vital; only materials planning.

10. MRP-II – Manufacturing Resource Planning

- **MRP II** = extension of MRP-I into a **closed-loop** system that plans **all the resources** of the company – materials, **capacity (machines, labour), finance, marketing** – and links with **simulation ("what-if")**. (Oliver Wight, 1980s.)
- Closed-loop MRP: MPS → MRP → **capacity requirements planning** → **shop floor control / purchasing** → **feedback** to the planning levels.



feedback (closed loop) from SFC to planning levels

Modules of MRP II

- **Business planning, sales and operations planning (aggregate planning)**, demand management & forecasting, **MPS**, **rough-cut capacity planning**, **MRP**, **capacity requirements planning**, **shop-floor control**, purchasing, inventory management, **financial and cost accounting**, distribution resource planning (DRP), performance measurement.

MRP-I vs MRP-II

MRP-I	MRP-II
Material planning only	Plans all the resources (material, capacity, money)
Open loop, infinite capacity assumed	Closed loop with capacity check and feedback
Output: order releases	Output: integrated plan with finance and marketing
Few modules, 1970s	Many modules; basis of ERP (1990s)

- **ERP** (Enterprise Resource Planning) extends MRP II to the whole enterprise: HR, finance, supply chain, CRM, etc. (SAP, Oracle).

11. CIMS – Computer Integrated Manufacturing Systems and their Benefits

- **CIMS** = the integrated system of **CAD, CAPP, CAM / CNC, FMS, CAQC, AS/RS, robots, MRP / PPC and business functions** connected by a **network and common database (LAN, MAP / TOP protocols)** – the "factory of the future".
- Levels (hierarchy): **plant / enterprise level → shop / area level → cell level → workstation level → equipment level**.

Benefits of CIMS

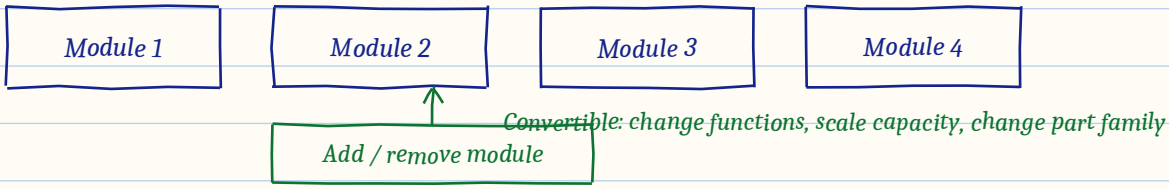
- **Shorter lead time**: 20 to 60% shorter design and production cycle; lower **inventory (WIP)** by 30 to 60%.
- Higher **productivity (40 to 70%)**, better utilisation of machines, less scrap and **better quality**, lower cost of products.
- **Flexibility** and quick response to customer demands; better **customer service**, low cost of changes.
- Better **management control** with real-time information, better decisions, lower indirect labour, no duplication of data.
- Competitive advantage in the global market. Disadvantages: very high investment, complexity, need for skilled people, compatibility of vendors' systems, resistance to change.

Trends in Manufacturing Systems

12. Reconfigurable Manufacturing

- **Reconfigurable manufacturing system (RMS)** (Y. Koren, 1999) = a system designed at the outset for **rapid change in structure** (hardware and software) so that it can quickly **adjust production capacity and functionality** in response to sudden market changes or product changes.
- It sits between dedicated lines (high volume, no flexibility) and FMS / CNC (high flexibility, lower throughput): RMS gives **customised flexibility** – only the flexibility needed for a part family.

Reconfigurable manufacturing system (RMS)



Six core characteristics of RMS

- **Modularity** – components (machines, spindles, fixtures, software) are modules that can be added / removed.
- **Integrability** – quick and precise integration of modules through standard interfaces.
- **Customisation** – flexibility limited to a part family (customised flexibility and control).
- **Convertibility** – ease of changing the functionality to new products (change spindle heads, tools).
- **Scalability** – capacity can be increased / decreased by adding or removing modules or stations.
- **Diagnosability** – quick identification of quality and reliability problems at the source.

RMS vs DML vs FMS

Feature	Dedicated (DML)	FMS	RMS
Flexibility	none	general	customised (part family)
Capacity change	fixed	fixed	scalable
Cost / throughput	low cost, high throughput	high cost, low throughput	moderate cost and throughput
Structure	fixed	fixed	changeable

- Benefits: faster launch of new products, lower response time to market changes, capacity matching demand, lower investment, better use of existing equipment (reuse of modules).

13. Sustainable Manufacturing

- **Sustainable manufacturing** = creation of manufactured products using processes that are **non-polluting, conserve energy and natural resources, and are economically sound and safe** for employees, communities and consumers (US Dept. of Commerce). Triple bottom line: **people – planet – profit**.
- Principles (**6R**): **Reduce, Reuse, Recycle, Recover, Redesign, Remanufacture**.
- Practices:
  - **Green design / design for environment (DfE)**: material selection (recyclable, non-toxic), design for disassembly, durable and repairable products, lightweight.
  - **Energy efficiency**: efficient motors, energy monitoring, regenerative drives, renewable energy, heat recovery.
  - **Resource efficiency**: near-net-shape processes (casting, **additive manufacturing**), less scrap, **MQL / dry machining** (minimum quantity lubrication replaces flood coolant), water recycling.
  - **Waste management**: zero-waste to landfill, **end-of-life** recycling and remanufacturing, **closed-loop supply chain** and **circular economy**.
  - **Life-cycle assessment (LCA)**: environmental impact from raw material → manufacturing → use → disposal; carbon footprint; ISO 14001 environmental management.
- Enablers: Industry 4.0 / IoT sensors for energy monitoring, digital twins, simulation

to cut trial-and-error.

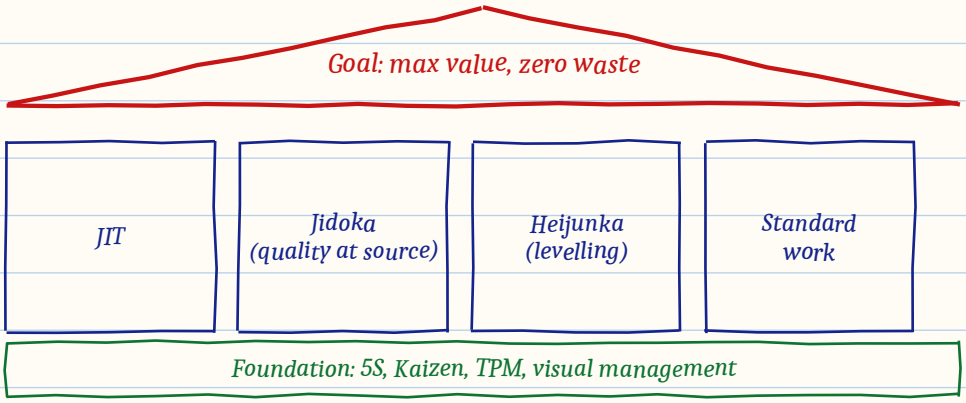
- Benefits: lower cost of energy and material, compliance with regulations, better brand image, safe working environment, long-term competitiveness. Challenges: initial cost, lack of data and standards, change of mind-set, supplier cooperation.

14. Lean Manufacturing

- **Lean manufacturing** (from the **Toyota Production System**, Taiichi Ohno) = a systematic approach to **eliminate waste (muda)** and create **maximum value for the customer** using the least resources.
- **Five lean principles** (Womack & Jones): (1) specify **value** from the customer's view, (2) identify the **value stream**, (3) create **flow**, (4) **pull** production, (5) pursue **perfection** (continuous improvement).

Seven wastes (TIMWOOD)

- **T**ransportation, **I**nventory, **M**otion, **W**aiting, **O**verproduction (worst), **O**ver-processing, **D**efects (+ 8th: unused creativity of people).



Lean tools and techniques

- **JIT (Just-In-Time)**: produce only what is needed, when needed, in the amount needed; **Kanban** (card signal system for pull); **Heijunka** (production levelling); **Takt time** = available time / customer demand.
- **Jidoka** (autonomation): machine stops automatically on a defect; **Poka-yoke** (mistake-proofing); **Andon** (alert system).
- **5S**: Sort (Seiri), Set in order (Seiton), Shine (Seiso), Standardise (Seiketsu), Sustain (Shitsuke).
- **Kaizen** (continuous improvement), **SMED** (single-minute exchange of die – quick changeover), **TPM** (total productive maintenance), **VSM** (value stream mapping), **cellular manufacturing / U-cells**, standard work, visual management.

- **Benefits**: lower inventory and lead time, better quality, less floor space, higher productivity, flexibility, employee involvement, lower cost.
- **Limitations / challenges**: needs stable demand and reliable suppliers, cultural change and training, vulnerable to disruptions (low buffers).

Lean, agile, sustainable – a comparison

Aspect	Lean	Reconfigurable	Sustainable
Focus	waste elimination, flow	quick change of capacity & function	environment, energy, resources
Tool	JIT, kaizen, 5S	modular machines	6R, LCA, green design
Result	low cost, high quality	responsiveness to market	long-term eco-efficiency

15. Quick Revision

CAPP: retrieval (variant, GT-based, edit standard plan) vs generative (logic, builds plan; CAD features).

- Machinability data: Taylor $VT^n = C$; min-cost vs max-production speed. Time standards: setup + machining + handling + allowance.
- Capacity planning: load vs available capacity; SFC: release – dispatch – monitor.
- MRP-I: MPS + BOM + inventory → planned orders; MRP-II: closed loop with capacity & finance; CIMS benefits.
- Trends: Reconfigurable (modularity, scalability, convertibility, integrability, customisation, diagnosability); Sustainable (6R, LCA); Lean (waste, JIT, kaizen, 5S, kanban).

Likely exam questions

- Explain retrieval and generative CAPP with block diagrams; compare them.
- What is a machinability data system? Explain Taylor's equation and its use.
- Explain computer generated time standards.
- Explain capacity planning and shop floor control.
- Explain MRP-I with inputs and outputs; differentiate between MRP-I and MRP-II.
- Write the benefits of CIMS.
- Explain reconfigurable, sustainable and lean manufacturing concepts.